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IT601 [Practical]

System and network Administration

LAB SOLUTIONS

Self Made Notes

Made by Mah jabeen

[mahjabeen97869@gmail.com]

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LAB 1

INSTALL UBUNTO AND VIRTUAL BOX WITH BASIC DEFAULT SETTINGS IN LAB 1

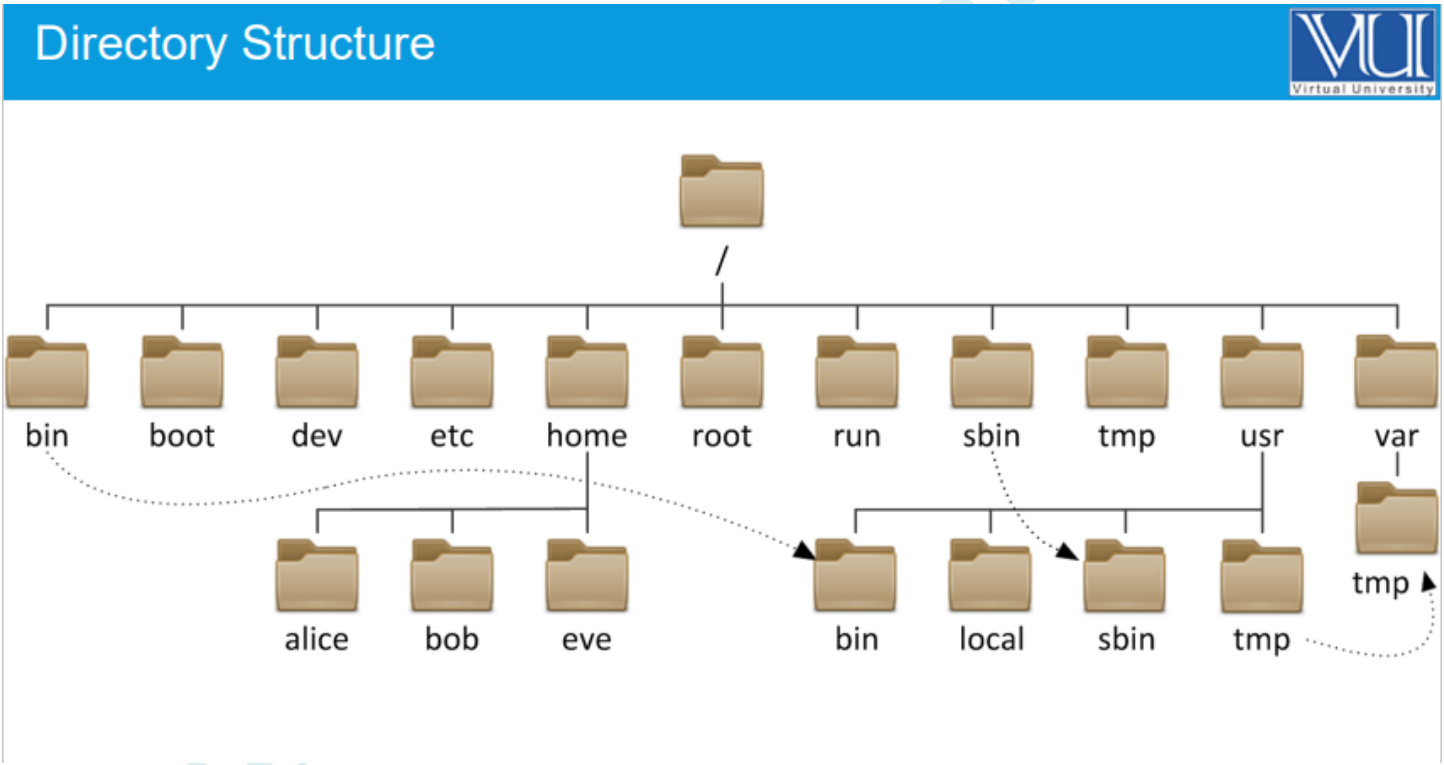
Contents

- Installing Linux Server
- Preparing Oracle Virtual Box
- Downloading Linux Server OS
- Setting up OS

mahjabeen97869@gmail.com

Walk Through Linux Server Basic Commands

- Creating folders and files
- Creating files using redirection
- Moving and manipulating files
- About Files
- The command line and the superuser
- Hidden files
- Mount points



/ — The Root Directory

/bin — Essential User Binaries

/boot — Static Boot Files

/cdrom — Historical Mount Point for CD-ROMs

/dev — Device Files

/etc — Configuration Files

/home — Home Folders

/lib — Essential Shared Libraries

/lost+found — Recovered Files

/media — Removable Media

/mnt — Temporary Mount Points

/opt — Optional Packages

/proc — Kernel & Process Files

/root — Root Home Directory

/run — Application State Files

/sbin — System Administration Binaries

/selinux — SELinux Virtual File System

/srv — Service Data /tmp — Temporary Files

/usr — User Binaries & Read-Only Data

/var — Variable Data Files

SUMMARY OF THESE POINTS

Linux Root Directory Strucuture

/ — **The Root Directory:** Contains all system files, similar to Windows' C:\ directory.

/bin — Essential User Binaries: Essential user binaries; /usr/bin holds applications like Firefox.

/boot — Static Boot Files : Stores boot-related files (GRUB, kernels); config files are in /etc.

/cdrom — Historical Mount Point for CD-ROMs: Historical CD-ROM mount point; /media is now standard for temporary media.

/dev — Device Files: Special files representing devices (e.g., /dev/sda for SATA drives).

/etc — Configuration Files: System-wide config files; user-specific configs are in home directories.

/home — Home Folders: User home folders (e.g., /home/bob) is for data and settings.

manjabeer97869@gmail.com

/lib — Essential Shared Libraries: The /lib directory contains libraries needed by the essential binaries in the /bin and /sbin folder. Libraries needed by the binaries in the /usr/bin folder are located in /usr/lib.

/lost+found — Recovered Files: it holds recovered files after file system crashes.

/media — Removable Media: Auto-mounts removable media devices (e.g., CDs).

/mnt — Temporary Mount Points: Historically used for temporary mounts; we can mount other file systems anywhere on the system.

/opt — Optional Packages: It **c**ontains optional software packages.

/proc — Kernel & Process Files : The /proc directory similar to the /dev directory because it doesn't contain standard files. It contains special files that represent system and process information.

/root — Root Home Directory: The /root directory is the home directory of the root user. Instead of being located at /home/root, it's located at /root. This is distinct from /, which is the system root directory.

/run — Application State Files:

The /run directory is fairly new, and gives applications a standard place to store transient files they require like sockets and process IDs. These files can't be stored in /tmp because files in /tmp may be deleted.

/sbin — System Administration Binaries: The /sbin directory is similar to the /bin directory. It contains essential binaries that are generally intended to be run by the root user for system administration.

/selinux — SELinux Virtual File System

If your Linux distribution uses SELinux for security (Fedora and Red Hat, for example), the /selinux directory contains special files used by SELinux. It's similar to /proc. Ubuntu doesn't use SELinux, so the presence of this folder on Ubuntu appears to be a bug.

/srv — Service Data

The /srv directory contains "data for services provided by the system." If you were using the Apache HTTP server to serve a website, you'd likely store your website's files in a directory inside the /srv directory.

/tmp — Temporary Files

Applications store temporary files in the /tmp directory. These files are generally deleted whenever your system is restarted and may be deleted at any time by utilities such as tmpwatch.

/usr — User Binaries & Read-Only Data: It contain user binaries like /usr directory contains applications and files used by users, as opposed to applications and files used by the system. For example, non-essential applications are located inside the /usr/bin directory ,non-essential system administration binaries are located in the /usr/sbin directory. Libraries for each are located inside the /usr/lib directory. Some other directories — for example, architecture-independent files like graphics are located in /usr/share. The /usr/local directory is where locally compiled applications install to by default — this prevents them from mucking up the rest of the system.

/var — Variable Data Files: The /var directory is the writable counterpart to the /usr directory

mohjabeen97869@gmail.com

LAB 02

Walk Through Linux Server Basic Commands

- ➢Location
- ➢ Creating folders and files
- ➢ Creating files using redirection
- ➢ Moving and manipulating files
- ➢ About Files
- ➢ The command line and the superuser
- ➢ Hidden files
- ➢ Mount points

Location

- pwd
- cd
- Relative path
- Absolute path

Basic Operation on Linux

how to create folders

1) current working path ko daikhy k Liya pwd ki command use Kary gy pwd print working directory
   - it601p@labserver : ~$ pwd
2) files or folders in current working directory we use dir or ls
   - it601p@labserver : ~$ dir (or) it601p@labserver : ~$ ls
3) To change the directory home to root
   - it601p@labserver : ~$ cd /
4) current dir sy ik step wapis jany k liya
   - it601p@labserver : ~$ cd ..
5) check the files on root
   - it601p@labserver : /$ ls
6) root direct sy 2 step back ja k etc directory Mai jany k liya

mahjabeen97869@gmail.com

- it601p@labserver : /$ ../../etc

7) To create directory lab 1 on home
   - it601p@labserver : ~$ mkdir lab1

8) ager name Mai space ho to
   - it601p@labserver : ~$ mkdir "lab 1"

9) lab 1 k Ander ja k 3 different folder bna sakty pahly cd lab1 kary gy :~$ change ho k ~/lab1$ ho jy ga then
   - it601p@labserver : ~/lab1$ mkdir ex1 ex2 ex3

10) root py rah k ham chaty k home Mai new lab2 create kar lain
   - it601p@labserver : /$ mkdir ~/lab2

11) ager ham chaty k wo directory again na bny Jo pahly sy bni ho
   - it601p@labserver : ~$ mkdir -p lab2

12) -p -v switch jab koi error ho to hmy verbally bta daita hy
   - it601p@labserver : ~$ mkdir -p -v lab1 lab2 lab3

HOW TO CREATE FILES

1) To create empty file in home
   - it601p@labserver : ~$ touch myfile.txt

2) ager is file k content daikhny ho
   - it601p@labserver : ~$ cat myfile.txt

3) ik command ki output ko dosry mai transfer karny k liya
   - it601p@labserver : ~$ ls > myfile.txt

4) simple string ko display karwana chaty
   - it601p@labserver : ~$ echo " this is test"

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5) Isi ko ager dosry Kisi folder Mai bajna chaty
   - it601p@labserver : ~$ echo "this is a test" > myfile.txt

6) ager hm chaty k replace na kry new likh dy to double >>
   - it601p@labserver : ~$ echo "this is a test" >> myfile.txt

7) ager file boht bari hy size Mai to ham less use Kary
   - it601p@labserver : ~$ less myfiles.txt

8) move files from one dir to other
   - it601p@labserver : ~$ mv myfiles.txt lab1

9) move all files from source directory to destination directory
   - it601p@labserver : ~$ *.*

10) Kisi khas pattern ko use kar k files move karna
   - it601p@labserver : ~$ mv mf*.* lab2

11) copy k liya
   - it601p@labserver : ~$ cp my file.txt lab1/

12) remove file
   - it601p@labserver : ~$ rm myfile.txt

13) remove directory
   - it601p@labserver : ~$ rmdir 'lab 4'

14) completely removed the directory
   - it601p@labserver : ~$ rm –r lab1

File ,word count etc wo sab daikhy gy ab

1) Sab files ko as a tree form mai daikhny k liya intall utility firstly Install pkg

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- it601p@labserver : ~$ sudo apt install tree

2) check the total number of lines of output file
   - it601p@labserver : ~$ wc -l output.txt

3) tree ko path btany k Kiya k kha sy build start karni
   - it601p@labserver: ~$ tree ~| wc – l

4) To count unique file in home folder
   - it601p@labserver: ~$ ls wc -l

5) tree k Ander home folder Mai unique files
   - it601p@labserver: ~$ tree ~ |Uniq |wc –l

6) To check files of etc directory on home
   - it601p@labserver: ~$ ls/etc to see less output
   - it601p@labserver: ~$ ls /etc/ less

7) To check the manual of unique command
   - it601p@labserver: ~$ man uniq

8) To sort the content of file
   - it601p@labserver: ~$ sort output.txt

9) To check k ham abi kis user k Tor py log in hy system mai
   - it601p@labserver: ~$ whoami

10) Root apt ki Kisi file ko remove karny k Liya e.g tree
   - it601p@labserver: ~$ sudo apt remove tree

11) To hide the file we use (.) dot
   - it601p@labserver: ~$ mv output.txt .output.txt

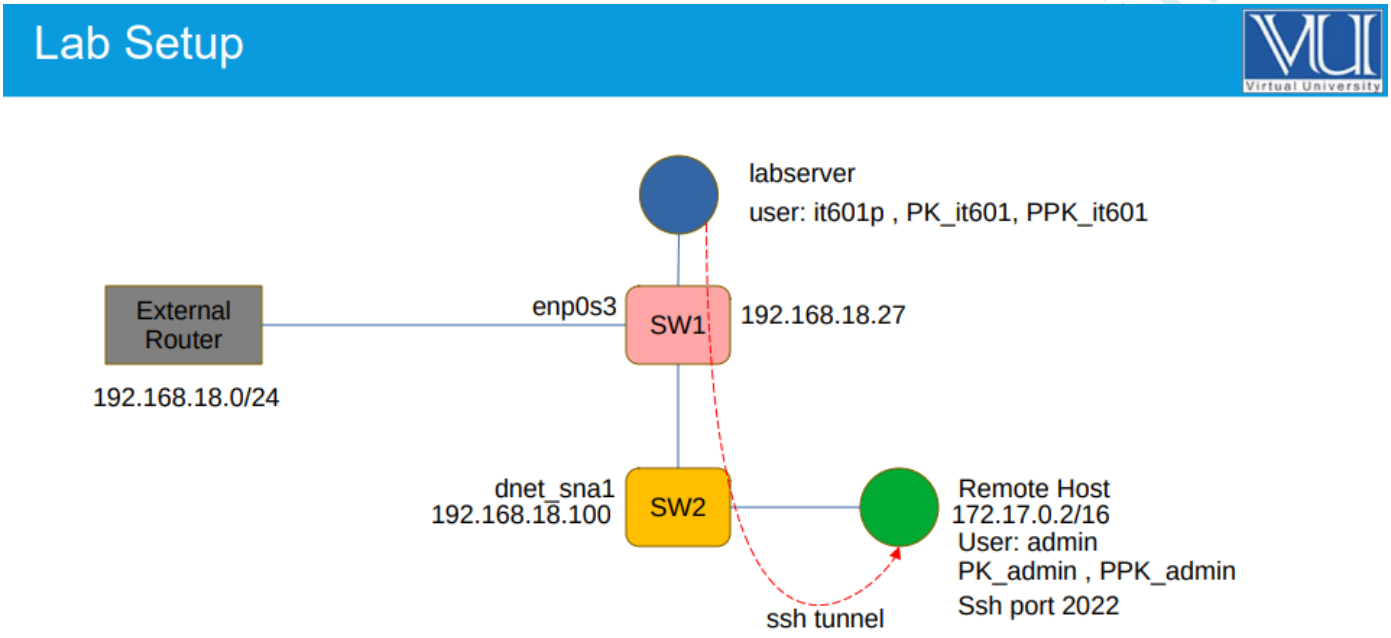
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12) hidden files ko seen karny k Liya dot . Laga k onko daikh sakty

- it601p@labserver: ~$ ls.output.txt
- it601p@labserver: ~$ ls -a.output.txt kar k b daikh sakty

LAB 3



SSH Access with Remote Username and Password

1. view Ip address information
   $ Ifconfig -a |more $ Ip a
2. Update APT repositories and install openssh-server
   $ sudo apt update
   $ sudo apt install openssh-server
3. Check status of the ssh server
   $ sudo systemctl status ssh
4. Check status current sshd configuration and default port to 2022
   $ sudo Vi /etc/ssh/sshd_conf
5. connect remote server with ssh on custom port and using remote user password
   $ ssh -p 2022 <remote Username>@<remoteipAddress>

SSH Access with Public Key (password less)

1. Generate RSA keys at labserver
   $ ssh-keygen -t rsa -b 4096
   #define passphrase
2. verify key pair
   $ Is !/.ssh/
3. copy public key of lab server user to remove server
   $ ssh-copy-id "-p <remotePort> <remoteUsername>@<remotelpAddress>"
4. connect remote server with ssh on custom port without remote user password
   $ ssh -p 2022 <remote Username>@<remotelpAddress>

How to enable remote access

Remote access ki utility ko enable karny k Kiya pahly ham ssh ki utility ko install Kary gy fr daikhy gy k ham kasy remote host ko access kar sakty

1. Firstly ya hamara virtual server hy jis ko ham ny lab server py create kiya tha or osko console k through access kiya Howa hy
   - admin@215e246fe109:/$

2. To install net tools because without net tools if config command doesn't work
   - admin@215e246fe109: /$ sudo apt install net-tools

3. To check the IP address and other interfaces
   - admin@215e246fe109: /$ ifconfig

1. Ping the IP address
   - admin@215e246fe109: /$ 172.17.0.2

2. ping the default gateway
   - admin@215e246fe109: /192.168.18.1

3. To install openssh-server

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- admin@215e246fe109 : /$ sudo apt install openssh-server

4. To check the status after installation (working or not)
   - admin@215e246fe109: /$ sudo systemctl status ssh

5. To check the configuration of ssh
   - admin@215e246fe109: /$ cd /etc/ssh/

6. To change the port 22 to port 2022
   - admin@215e246fe109: /etc/ssh$ sudo vi sshd_config

7. To restart the ssh service
   - admin@215e246fe109: /etc/ssh$ sudo systemctl restart ssh

sudo k bggair siraf daikh sky gy change ni kr saky gy port i is used for uncomment

Go back to original lab server

- it601p@labserver: ~$

8. ping through original server
   - it601p@labserver: ~$ ping 172.17.0.2

9. To access the admin user through ssh
   - it601p@labserver: ~$ ssh <username> @ < IP address>
   - it601p@labserver: ~$ ssh -p 2022 admin@172.17.0.2

Ab ham daikhy gy k ham ssh ko public and private keys k access kar sakty ta k Hamy password ki zarorat na oar k zariya kasy

10. generate keygen packages to enable keys

mahjabeen97869@gmail.com

- it601p@labserver: ~$ ssh-keygen -t rsa -b 4096

11. To check the ssh directory
    - it601p@labserver: ~$ ls.ssh/

12. transfer the public key to remote server
    - it601p@labserver: ~$ ssh-copy-id -p 2022 admin@172.17.0.2

13. To check the authorized keys available or not
    - admin@215e246fe109: /$ ls -a ~/. Ssh

LAB 04

User management on Linux server Orr root user ko b daikhy gy and user ko add remove karny k bary mai b daikhy gy

Normally root user is locked in Ubuntu osko khud enable Karna parta

1. To change password
    - it601p@labserver: ~$ sudo passwd

2. To view all user on system
    - it601p@labserver: ~$ less /etc/passwd

3. To check the password of root is enabled or not
    - it601p@labserver: ~$ grep root /etc/passwd

for more information

- it601p@labserver: ~$ sudo grep root /etc/shadow

mahjabeen97869@gmail.com

4. To use the cut utility
   - it601p@labserver: ~$ cut-d: -f 1/etc/passwd

5. To see the group information
   - it601p@labserver: ~$ less/etc/group

6. To check the name of groups /etc/group
   - it601p@labserver: ~$ cut -d: - f 1/etc/passwd

7. To check the currently login user
   - it601p@labserver: ~$w

8. To check the manual of sudo
   - it601p@labserver: ~$ man sudo

8. Add user
   - it601p@labserver: ~$ sudo adduser student1

9. To get the information of student1
   - it601p@labserver: ~$ sudo grep student1 /etc/passwd

10. delete user
    - it601p@labserver: ~$ sudo deluser student1

11. To add group
    - it601p@labserver: ~$ sudo addgroup vustudents

12. Add student into group
    - it601p@labserver: ~$ sudo adduser student1 vustudents

mahjabeen97869@gmail.com

13. To check the permission of user
    - it601p@labserver: ~$ ls -ld /home/student1

drwxr mean globally readable permissions

0750 mean less permissions

14. To change the permission k other user information ko view na kar saky
    - it601p@labserver: ~$ sudo chmod 0750 /home/student1

15. To check the conf permission information
    - it601p@labserver: ~$ sudo vi /etc/adduser.conf

16. To check the information about password
    - it601p@labserver: ~$ sudo vi /etc/pam.d/common.password

17. To check the expiry of current user password
    - it601p@labserver: ~$ sudo chage -l it601p

After deleting the user you should delete the home directory of deleted user

LAB 05

how to use containers to deploy different services

container are light weighted but virtual machines need extra resources container share resources on run time

mahjabeen97869@gmail.com

1. To check docker package is installed Orr not
   - it601p@labserver: ~$ dpkg -1 | grep docker*

2. remove docker
   - it601p@labserver: ~$ sudo apt remove docker -*

3. dockerk related all files/setting ko del karny k liya
   - it601p@labserver: ~$ sudo purge docker*

we required three types of dependencies

1) cs-certificates

2) curl

3) gnupg

There are two types keys

1) public key

2) private key

Inko ham authentication or encryption k Liya use kar sakty

Ca certificates are electronic passwords jinko ham authentication and or encryption k Liya use karty hy

ca certificates k Liya Jo certified authority hoti hy os k public certificates ko manage karny k Liya ca-certificate ka package use hota hy isko different server communication ya identity verification k Kiya b use kar sakty hy

4. install ca-certificate
   - it601p@labserver: ~$ sudo apt install ca-certificate

mahjabeen97869@gmail.com

curl package is a client url Jo k diff web server k sath communication k Liya use hota hy eg file download ya server sy response retrieve karny k liya

5. install curl package
   - it601p@labserver: ~$ sudo apt install curl

next package is pretty good privacy gnu pgp isko ham different files ko encrypt karny k Liya use Kary gy ya fr hmari communication ko sign ya is sy ham apni or remote server ki identity ko verify kar saky gy isko ham ssh k sath b use kar sakty

6. install gpg
   - it601p@labserver: ~$ sudo apt install gnupg

These are some of our initial requirements before installing docker

Now we have to install the keys of docker in our system

0755 mean world/ globally readable

7. To change the permission of directory
   - it601p@labserver: ~$ sudo install-m 0755 -Id/etc/apt/keyrings

-rw-R--R- has binary 0110100100-0755

8. force fully remove the gpg
   - it601p@labserver:~$ sudo rm -f /etc/apt/keyrings/docker.gpg

There are different types of switches in curl manual

-f for fail (fail silently mean no output generate)

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-s silent

-S show errors

-L it is for location option

9. Download docker with all these switches
   - it601p@labserver: ~$ curl -f -s -S-L https://download. docker.com/ Linux/Ubuntu/gpg

Download docker with all these switches and enable dearmore option it601p@labserver: ~$ curl -f -s -S-L https://download.docker.com/ Linux/Ubuntu/gpg | sudo-dearmore-o /etc/apt/keyrings/docker.com

10. check the read write permission of keyrings
    - it601p@labserver r : ~ S sudo chmod sudo chmod a+r/etc/apt/keyrings/ docker.gpg

now we have to repositories of docker and we have to a docker list in apt list now we have to build the architecture of current system

11. To set the repositories of docker
    - it601p@labserver: ~$ echo "deb [arch ="$ (dbkg --print-architechture) " sign by =/etc/apt/keyrings/docker.gog] https://download.docker.com/Linux/Ubuntu "$(./etc/os-release && echo "$VERSION_CODENAME")" stable tee/etc/apt/sources.list.d/docker.list >/dev/null

it601p@labserver:~$ echo "deb [arch="$(dpkg --print-architecture)" signed-by=/etc/apt/keyrings/docke

r.gpg] https://download.docker.com/linux/ubuntu "$(./etc/os-release & & echo "$VERSION_CODENAME")" s table" | sudo tee /etc/apt/sources.list.d/docker.list > /dev/null

echo "deb " \ #package-type

[arch= "$(dpkg --print-architecture)" \ #architecture = amd64

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signed-by=/etc/apt/keyrings/docker.gpg] \ #gpg certificate

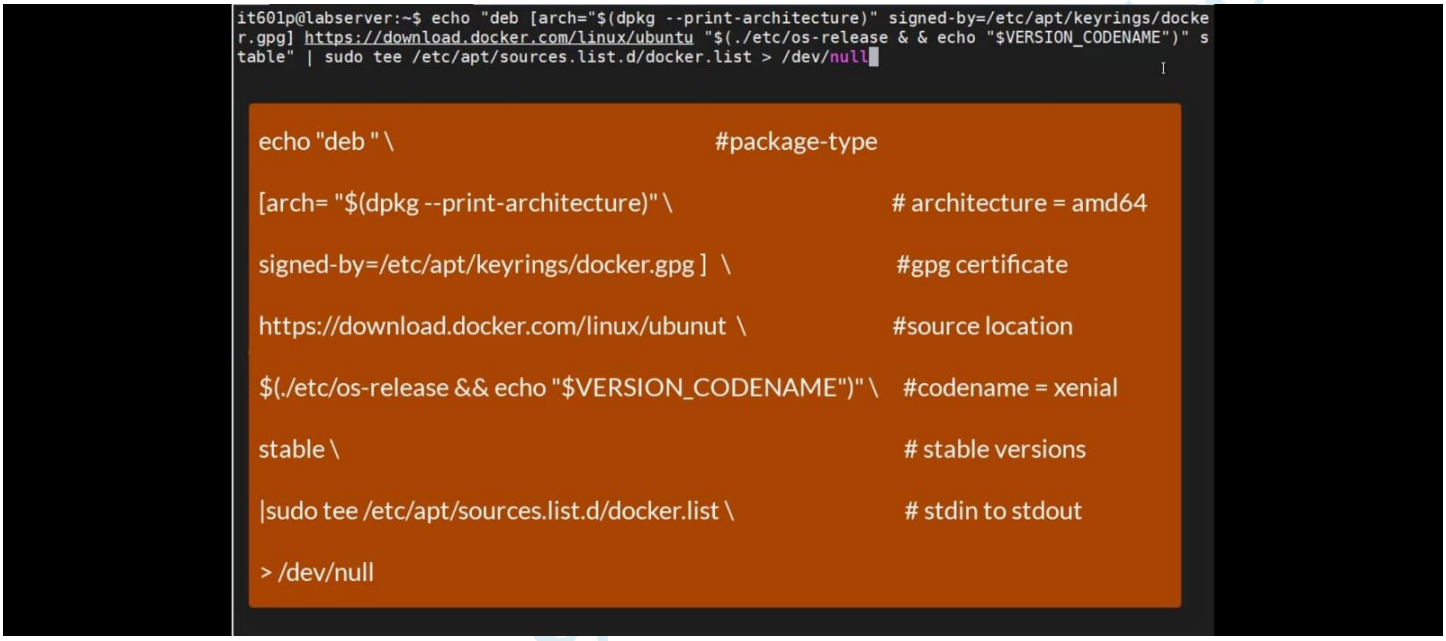
https://download.docker.com/linux/ubunut \ #source location

$(./etc/os-release && echo "$VERSION_CODENAME")" \ #codename = xenial

stable \ # stable versions

sudo tee /etc/apt/sources.list.d/docker.list \ # stdin to stdout

> /dev/null



12. To check these repositories set correctly or not
    - it601p@labserver: ~$ ls/etc/apt/sources.list.d

13. To view entries
    - it601p@labserver: ~$ cat /etc/apt/sources.list.d/docker.list

Now isko update kar dy gy by using sudo apt update

14. Now install different docker packages
    - it601p@labserver:~$ sudo apt-get install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin

15. To check that docker successfuly install or not

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- it601p@labserver: ~$ sudo run hello-world

LAB 06

Docker Image:

A Docker Image is a read-only file with a bunch of instructions. When these instructions are executed, it creates a Docker container.

Dockerfile:

Dockerfile is a simple text file that consists of instructions to build Docker images.

Syntax

comments

command argument argument1...

from mean wo Konsa os hy Konsa version hy jiski base py ya image built kiya jyy ga

Docker Commands for Creating a Dockerfile container

FROM - Creates a layer from the ubuntu:18.04

- FROM ubuntu:18.04

pull ke keyword se Ham files ko docker repository se apni image ke andar add kar sakte hain

PULL - Adds files from your Docker repository

- PULL. /file RUN - Builds your container RUN make /file

CMD mtlb run ke zariye Ham Apne image per standard Linux ki commands ko execute kar sakte hain eg copy,move,edit etc

Jab container run karega to wo kon si file ko run karega mean execute kry ga

CMD - Specifies what command to run within the

- CMD python /file/file.py

ENTRYPOINT

• allows specifying a command along with the parameters

• Syntax, ENTRYPOINT application "arg, arg1".

• Example, ENTRYPOINT echo "Hello, $name".

mahjabeen97...9@gmail.com

ADD

• ADD command helps in copying data into a Docker image

• Syntax, ADD /[source]/[destination]

• Example, ADD /root_folder/test_folder

ENV

• ENV provides default values for variables that can be accessed within the container

• Syntax, ENV key value

• Example, ENV value_1

MAINTAINER

• MAINTAINER declares the author field of the images

• Syntax, MAINTAINER [name]

• Example, MAINTAINER author_name

Docker Commands

Name : Description

1. Attach :Attach local standard input, output, and error streams to a running container
2. Build: Build an image from a Dockerfile
3. Commit: Create a new image from a container's changes
4. Config: Manage Swarm configs.
5. Container: Manage containers
6. Cp: Copy files/folders between a container and the local filesystem
7. Create: Create a new container
8. Exec: Execute a command in a running container
9. Export: Export a container's filesystem as a tar archive

mahjabeen97869@gmail.com

10. Image: Manage images
11. Images: List images
12. inspect: Return low-level information on Docker objects
13. kill: Kill one or more running containers
14. load: Load an image from a tar archive or STDIN
15. network: Manage network
16. port: List port mappings or a specific mapping for the container
17. ps: List containers
18. rename: Rename a container
19. restart: Restart one or more containers
20. rm: Remove one or more containers
21. rmi: Remove one or more images
22. run: Create and run a new container from an image
23. save: Save one or more images to a tar archive (streamed to STDOUT by default)
24. stats: Display a live stream of container(s) resource usage statistics
25. stop: Stop one or more running containers
26. top: Display the running processes of a container
27. volume: Manage volumes

Build a Docker Image with Dockerfile

Create a directory where the dockerfile will be stored.

- mkdir c1 && cd c1

Create dockerfile

- sudo vi dockerfile

####Contents of Dockerfile####

FROM ubuntu:22.04

MAINTAINER Virtual_University_of_Pakistan

ENV DEBIAN_FRONTEND noninteractive

RUN apt-get update && apt-get install -y ubuntu-server

EXPOSE 2022

#expose the port 2022

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CMD ["echo", "Welcome to it601p"]

#above command will print the message and container will exit.

#To keep it running run bash and keep it running

CMD ["PROGRAM_TO_RUN"]

Build a docker image with dockerfile

➢Verify the image is crated

- docker images

➢ Run an image

- docker run name:tag <options>

--workdir , -w	Working directory inside the container
--hostname , -h	Container host name
--interactive , -i	Keep STDIN open even if not attached
--ip	IPv4 address (e.g., 172.30.100.104)
--label , -l	Set meta data on a container
--link	Add link to another container
--mount	Attach a filesystem mount to the container
--name	Assign a name to the container
--network	Connect a container to a network
--rm	Automatically remove the container when it exits
--mac-address	Container MAC address (e.g., 92:d0:c6:0a:29:33)
--memory , -m	Memory limit
--volume , -v	Bind mount a volume

mahjabeen97869@gmail.com

Create a New Container

Now, create a Docker container from the Docker image we created in the previous step.

docker run --name it601p_m1

Build a Docker Image with Dockerfile

➢ Build the image using above docker file

- docker build --help
- docker build <options>

➢ Some Options

--file, -f : docker file name

--label : set meta data for image

--quite -q : suppress output

--rm : Remove intermediate containers

--tag, -t : set name or tag

--add-host : Add custom host to IP mapping

--network : set the networking mode for run commands

➢ Upon completion , it returns the image id

These are according to lab of week 06

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[mahjabeen97869@gmail.com]

mahjabeen97869@gmail.com

Short notes of lab 6

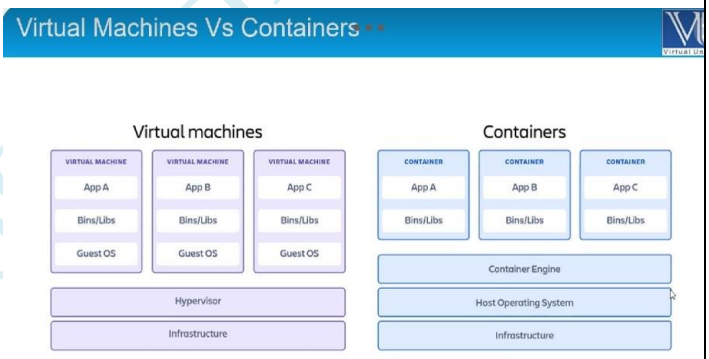
How to create container, different commands of docker, image creation

1. To check docker is working or no
   - it601p@labserver : ~$ docker --help

Difference between virtual machine and containers

virtual machine mai computing resources ko divide kar daity hy different vm Mai but container Mai ham different resources ko allocate b kar sakty or shere b kar sakty hain

1. firstly make c1 directory
   - it601p@labserver : ~$ mkdir c1



2. now create docker file in c1
   - it601p@labserver: ~/c1$ sudo vi Dockerfile

is used to add comment

3. Add comment in dockerfile
   - First image, Name: c1

4. Specify the os and it's version
   - FROM Ubuntu:22.04

5. Specify the Author information
   - MAINTAINER "virtual university of Pakistan"
   - we can also add our vu id here

6. To make the front end non interactive

mallabeen9786@gmail.com

- ENV DEBIAN_FRONTEND noninteractive

❖ ham ny os py Jo image Bnai os py koi command execute Karna chaty firstly update Kary gy repositories ko fr package ko install kar dy ga

- RUN apt-get update && apt-get install -y Ubuntu-server

7. container ki tcp /IP ports ko expose karny k Liya ta k host sy b on port ko access kar saky
   - EXPOSE 2022 chahm

8. hmary pass option hota k ham konsi command execute karna chaty or konsi nai
   - CMD["bash"]

To save all these setting we type wq and then exit from Dockerfile

9. Check the content of Dockerfile
   - it601p@labserver: ~/ c1$ cat Dockerfile

To create the container we have to build the image of this Dockerfile in the directory where we create the Dockerfile .

10. Build image of Dockerfile and specify it's name
    - it601p@labserver: ~/c1$ sudo docker build -t it601p:c1.

11. To check image build or not
    - it601p@labserver: ~/c1$ sudo docker images

12. To create the container we can run it and then detach it without directly connect to make it interactive
    - it601p@labserver: ~/c1$ sudo docker run -i -t it601p:c1 bash

After running this command we can directly go to system container that is root@143698f994d3:/#

mahjabeen97869@gmail.com

13. To see the directory of container
    - root@143698f994d3:/# dir

14. Go to home directory of container
    - root@143698f994d3:/# cd home/

15. Go to rootdirectory of container
    - root@143698f994d3:/# cd root

16. Install net tool to run ifconfig
    - root@143698f994d3:/# apt install net-tools

17. To check the IP address
    - root@143698f994d3:/# ifconfig

18. To check the container information from our host operating system like is time konsa container run kar raha hy
    - it601p@labserver: ~$ sudo docker ps

Ab jasy k hm ny net tools ka package kiya hy install to ager commit nai kry gy to wo delete ho jyy ga

19. To commit mean save the operation on container
    - it601p@labserver: ~$ sudo docker commit 143698f994d3

➤This is our container id 143698f994d3

20. To stop the container
    - it601p@labserver: ~$ sudo docker stop 143698f994d3

mahjabeen97869@gmail.com

21. To start the container again after stopping
    - it601p@labserver: ~$ sudo docker start 143698f994d3 –i

LAB 07

AJ ham services k part ko discuss Kary gy k kasy different services ko

Linux py deploy kar sakty

firstly ham DHCP service ko daikhy gy it is dynamic host configuration protocol

hy Jo ik lan py different host and servers ko ip address assign karta hy

ya addresses dynamic b ho sakty and static b randomly b or sequential b

we use two computer machines

1) DHCP server

2)client (web server)

1. To run the DHCP server firstly update the repositories
    - root@f9b0a70805f:/# apt install isc-dhcp-server

2. To check the manualy configured IP address
    - root@f9b0a70805f:/# ifconfig

3. etc folder Mai etc/dhcp/dhcpd.conf ko change Karna ho ga
    - root@f9b0a70805f:/# vi /etc/dhcp/dhcpd.conf

    - default least time 600sec

    - max least time 200sec
    - ➢Subnet
    - jis subnet py Kam Karna hy

```
default-lease-time 600;
max-lease-time 200;
authoritative;

subnet 192.168.19.0 netmask 255.255.255.0 {
  range 192.168.19.100 192.168.19.200;
  option  routers 192.168.18.1;
  option domain-name-servers 192.168.18.1, 8.8.8.8,8.8.4.4;
  #option domain-name "it601.vu.edu.pk";

}

host archmachine {
hardware ethernet 76:2d:54:fd:46:80;
fixed-address 192.168.19.3;
}
```

mahjabeen97869@gmail.com

➢ Range

mean wo kon c IP address hy Jo ham assisign Karna chaty

➢Router gateway address

next ya define kry gy k Jin host ko ip address assign Kary gy onko gateway Konsa assign kiya jyy ga

➢DNS server b define kar sakty and optional domain b

4. To assign the IP address statically

To assign the IP address statically we need need Mac address of our particular host or server Yani Ethernet interface jis sy wo connected hy os ka Mac address chaiya ho ga os ki base py static entries define kar sakty

For this purpose we have host key word and oska name fr next hardware address then oska fixed address

5. To change the netplan configuration and enable DHCP
   - root@f9b0a70805f:/ etc/netplan#

6. To change the range then run this command and change the range manually
   - root@f9b0a70805f:/# vi /etc/dhcp/dhcpd.conf

7. After changing restart the system
   - root@f9b0a70805f:/# systemctl restart isc-dhcp-server.service

8. To check the status after restart the system
   - root@f9b0a70805f:/#systemctl status isc-dhcp-server.service

9. To check the netplan setting
   - root@f9b0a70805f:/etc/netplan# vi 01-netcfg-yaml

10. To get the address
    - root@f9b0a70805f://etc/netplan# netplan apply

and

- root@f9b0a70805f:/#/etc/netplan# ifconfig

mahjabeen97869@gmail.com

Part 2

web services are use to communicate information on internet is ka maqsad ya hota k service provider information public k sath shere kar saky

AJ is lab mai ham web services ko daikhy gy

➢ on window os we use

Microsoft internet information services (IIS)

➢On Linux OS we use

Apache server

{Apace 2 is latest version of apache server}

In this lab we firstly Install Apache server with it's latest version apache2 fr ham default web service create Kary gy fr osko external host sy access Kary gy fr ham daikhy gy kasy ham multiple web services ik hi server py run kar sakty hy fr in services ko secure karny k bary mai daikhy gy

1. To intall the apache web server firstly update the repositories by apt update then install the apache package
   - root@web-server:/# apt install apache2

2. To check the status of this package
   - root@web-server:/# systemctl status apache2

at this this time service is loaded but it is dead mean inactive

3. To restart the service with default package then restart it by systemctl restart apache2 after that again check the status
   - root@web-server:/# systemctl status apache2

4. To check the service run locally on this system or not for this purpose we use curl.
   if curl is not install then install it
   - root@web-server:/# apt install curl

5. After that check curl on local host

maniabeen97869@gmail.com

- root@web-server:/# curl localhost

now this service is running

6. To check that this service is available on external host or not for this purpose we browse the IP address of external host if it show the message not reach mean it is not accessible

7. To check the IP address of external host we open the command prompt and write ipconfig command
   - PS C:\WINDOWS\System32>ipconfig
1. Add route on local host for 19.0 network k Liya for this purpose open CMD(command prompt) on laptop to fr hmary pass administrator mode Mai command prompt khul jayy ga
   - PS C:\WINDOWS\System32>

2. To add route on local host
   - PS C:\WINDOWS\System32>route add 192.168.19.0 mask 255.255.255.0 192.168.18.34 this is default gateway

This is default configuration

➢we have two important folders

1 - Webservice Contents on/var/www/html (is mai wo content hoty hy jinko ham present karty hy jab koi request Internet sy ati hy)

2 - Webservice Configurations on/etc/apache2 (ya configuration ki directory hy Jo etc folder Mai mojod hoti hy)

3. To check the content present on default path firstly go to www folder
   - root@web-server:/# cd /var/www/

4. list them by using ls then we saw html folder than go to html folder
   - root@web-server:/var/www#cd html/

5. To check or create the index .html
   - root@web-server:/var/www/html# vi Index.html

mahjabeen97869@gmail.com

6. To create backup copy of index file
    - root@web-server:/var/www/html#mv index.html index.html.bk

Basic format of html include these tags(html,head,boady)

1. To change the default content and give the content to external host

<html>

<header>

<title>

Virtual University of Pakistan

</title>

</header>

<body>

<h1>Welcome to default system and network administation page!/h1

</body>

</html>

2. To run multiple web services on same server we use apache it have virtual host module
    - root@web-server:/var/www/html#

3. To check the default configuration of apache
    - root@web-server:/var/www/html# cd /etc/apache2

4. To check the port information k konsi port py server run kar rha

mahjabeen97869@gmail.com

- root@web-server:/etc/apache2# vi ports.conf

hmy pta lgy ga k hmara server kis port py run kar rha by default wo port 80 py request send karta hy

5. To check all available sites k konsi configuration available Hain
   - root@web-server:/etc/apache2# ls sites-available/

6. To view it's contents firstly go to available sites
   - root@web-server:/etc/apache2# cd sites-available/
7. To view default configurations
   - root@web-server:/etc/apache2/sites-available# vi 000-default.conf

is sy Hamy pata lag jyy ga k ik hi virtual Host available hy jiska content port 80 py accessible hy

8. To change the name of default config
   - root@web-server:/etc/apache2/sites-available# vi 000-default.conf it601.conf
   - root@web-server:/etc/apache2/sites-available# vi 000-default.conf it601p.conf

9. To check the document root folder
   - root@web-server:/etc/apache2/sites-available# cd /var/www/

10. To create 2 directories with name it601 and it601p
    - root@web-server:/var/www/#mkdir it601
    - root@web-server:/var/www/#mkdir it601p

11. To change ownership and give their ownership to root
    - root@web-server:/var/www/#chown -R $USER:$USER it601
    - root@web-server:/var/www/#chown -R $USER:$USER it601p

12. To give the world readable permissions to www directory
    - root@web-server:/var# chmod -R 755 www/

mahjabeen97869@gmail.com

13. To copy the index.html file in it601and it601p
    - root@web-server:/var/www/# cp html/index.html it601/
    - root@web-server:/var/www/# cp html/index.html it601p/

we can go to both of these directories by cd command after that

- for going on index .html page we type
- vi index.html

14. To enable the configuration site of it601
    - root@web-server:/etc/apache2/sites-available# a2ensite it601.conf

15. To disable the configuration site
    - root@web-server:/etc/apache2/sites-available#a2dissite it601.conf

16. To enable the SSL
    - root@web-server:/etc/apache2/sites-available# a2enmode SSL

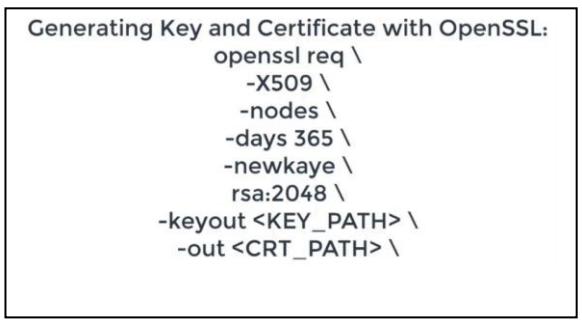
17. To get the certificate for https and SSL
    - root@web-server:/etc/apache2/sites-available# open ssl

18. To create the certificate with open ssl
    - root@web-server:/etc/apache2/sites-available# openssl req -X509 -nodes -days 365 -newkey rsa:2048-keyout /etc/ssl/private/apache-selfsigned.key -out /etc/ssl/certs/apache-selfsigned.crt

509 is certificate type 2048 is key size

Root path ki files ko browse b kiya ja Sakta hy is ko change kary gy because root files ko accessible nai hona chiyaa

To change the configuration firstly go to root directory



mahabeen97869@gmail.com

19. To create file .htaccess

- root@web-server:/etc/apache2/sites-available#vi .htaccess

than is Mai ja k option- indexes kar k save kar dain gy

20. To change the configuration by default

- root@web-server:/etc/apache2# vi apache2.conf

Ab ider ja k indexes ko remove kar dain gy fr system restart kar dain gy is fr root path ki files browse hona band ho jyy gi.

LAB 08

DNS Service

1.bind9

2.bind9utils

3.bind9-doc

Packages to install

Sudo apt aupdate && Sudo apt install bind9 bind9utils bind9-doc

1. Local Networks : 192.168.19.0/24

2. DNS SERVER: 192.168.19.2 , bindserver

3. WEB_SERVER : 192.168.19.3 , web

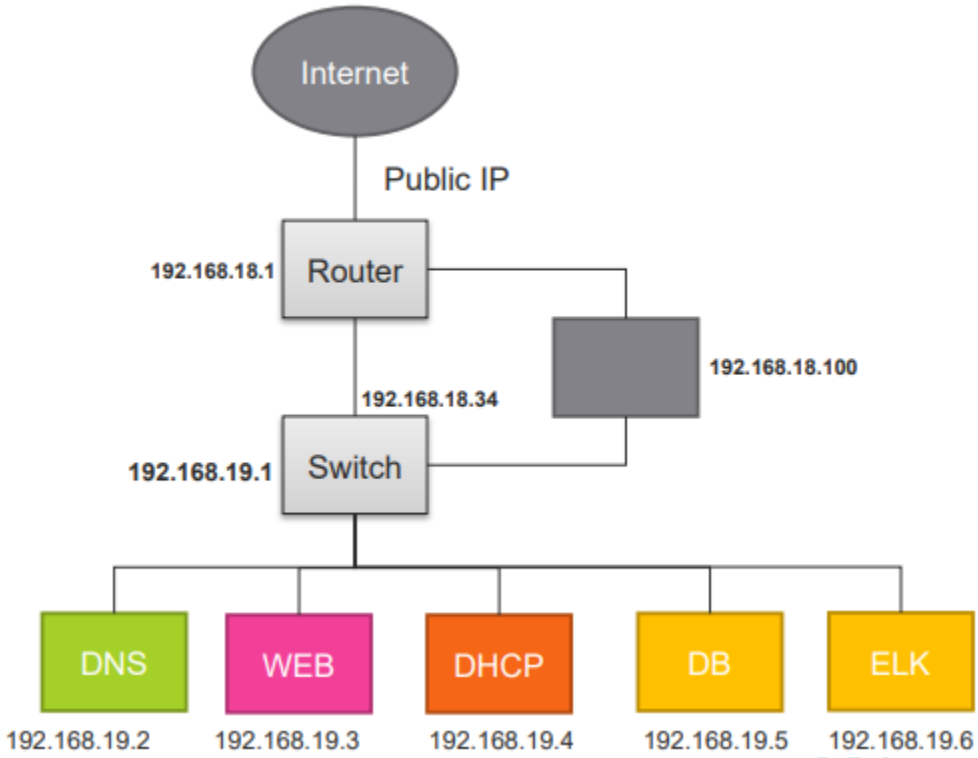
4. DHCP_SERVER : 192.168.19.4 , dhcp

5. DB_SERVER : 192.168.19.5 , db

6. ELK_SERVER : 192.168.19.6 , elk

7. DOMAIN : vusna.com

mahjabeen97869@gmail.com



Short notes of lab 8

DNS ham is Liya use karty hy k ham pory pory ip address ko yad rakhny k bachy Kisi common name sy IP addresses ko yad Rakh saky

common name type Kary to koi particular IP address ko refer Kary eg DNS snad for domain name server like

vu .edu.pk

To configure the DNS we have to install some packages Jin mai ik bind ka package hy jiska new version bind 9 hy because it is more stable firstly update the repositories

1. To install bind9 ,bind9 utilities, bind9 documentation
   - root@dns-server:/# apt install bind9 bind9utils bind9-doc -y

2. To check the system status
   - root@dns-server:/#systemctl status bind9

3. To start the bind service because it is I active

mahjabeen97869@gmail.com

- root@dns-server:/# systemctl start bind9

4. To check the IP address of current server
   - root@dns-server:/#ifconfig

5. To check the configuration files
   - root@dns-server:/# cd /etc/bind

6. To check and open the name file and check the old configuration
   - root@dns-server:/etc/bind/#vi named.conf.options

7. To define restrict list k konsy host traffic ko access kar saky
   - for this purpose we use Access List forwardder contain default gateway in case packet is not send to required host

```
//access-list
acl LAN {
192.168.0.0/16;
};
options {
directory "/var/cache/bind";
allow-query {localhost; LAN;};
forwarders {192.168.18.1;};
recursion yes;
};
```

8. To check that this configuration is valid or not
   - root@dns-server:/etc/bind# named-checkconf /etc/bind/named.conf.options

mahjabeen97869@gmail.com

9. To set the configuration of local dns
    - root@dns-server:/etc/bind# vi named.conf.local

firstly zone define kry gy fr os k Ander type define Kary gy fr file configuration kar dy gy type k bad next reverse zone define kry gy Jo k 0.116.10.10.in-addr.arpa hy ya basically Ethernet address hy

10. To check that this configuration is valid or not
    - root@dns-server:/etc/bind# named-checkconf /etc/bind/named.conf.options

```
zone "vusna.com" IN {

type master;

file "/etc/bind/zones/vusna.com";

};

zone "0.116.10.in-addr.arpa" IN {


type master;

file "/etc/bind/zones/vusna.com.rev"

};
```

11. To create the directory of zones
    - root@dns-server:/etc/bind# mkdir zones

then go to this directory by cd zones

12. To copy the db.local file from previous directory to current zones directory and change its name to vusna.com
    - root@dns-server:/etc/bind/zones# cp ../db.local ./vusna.com

mahjabeen97869@gmail.com

13. To edit this file and add some configuration

- root@dns-server:/etc/bind/zones# vi vusna.com

vusna.com. root.vusna.com. (

2 ; Serial

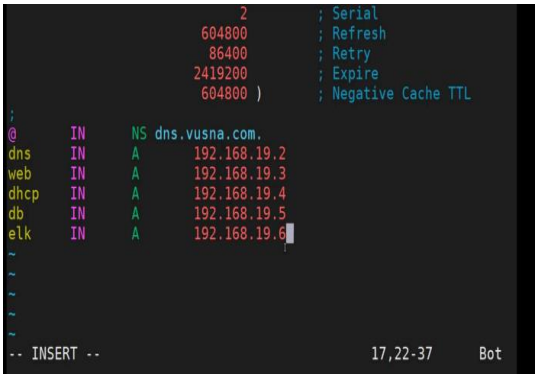
604800 ; Refresh

86400 ; Retry

2419200 ; Expire

604800) ; Negative Cache TTL

@ dns web dhcp db elk



VUSNA.COM Zone Configuration

$TTL 604800

@ IN SOA vusna.com. root.vusna.com. (3; 604800; 86400; 2419200; 604800) ;

@ IN NS bindserver.vusna.com.

bindserver IN A 192.168.2.2

web IN A 192.168.2.3

dhcp IN A 192.168.2.4

db IN A 192.168.2.5

elk IN A 192.168.2.6

14. To check the configuration is valid or not

- root@dns-server:/etc/bind/zones# named-checkzone vusna.com vusna.com

15. To configure the reverse zone firstly copy the kry reverse zone ki file ko fr osko edit kar ly gy

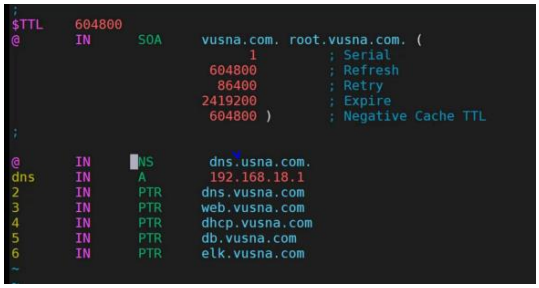
- root@dns-server:/etc/bind/zones# cp ../db.127 ./vusna.com.rev

marjabeen97860@gmail.com

16. To edit the reverse zone

- root@dns-server:/etc/bind/zones# vi vusna.com.reverse

Baki sab same ho ga local host ka name vusna.com kar dy gy



VUSNA.COM Reverse Zone Configuration

$TTL 604800

@ IN SOA vusna.com. root.vusna.com. (

2 ;

604800 ;

86400 ;

2419200 ;

604800) ;

@ IN NS bindserver.vusna.com.

bindserver IN A 192.168.18.1

2 IN PTR bindserver.vusna.com

3 IN PTR web.vusna.com

4 IN PTR dhcp.vusna.com

5 IN PTR db.vusna.com

6 IN PTR elk.vusna.com

mahjabeen97869@gma

17. To check the configuration is valid or not

- root@dns-server:/etc/bind/zones# named-checkzone vusna.com vusna.com.rev

reverse Mai ham sary ptr records ko daikhty hy

18. To check the DNS server configuration
    - root@dns-server:/etc/bind/zones# systemctl bind9

19. To check k Jo A record ya ptr ya phir Jo b host ham ny add kiyy hy onko common name sy access kar sakty ya nai
    - root@dns-server:/etc/bind/zones# ping web.vusna.com

1. To verify the DNS services by using nslookup it is the part of DNS utilities

firstly update the repositories

2. then install DNS utilities
    - root@dns-server:/etc/bind/zones# apt install dnsutils
    - root@dns-server:/etc/bind/zones# nslookup web.vusna.com

3. Asy hi ham apny DHCP host ki configuration ko b daikh sakty hy
    - root@dns-server:/etc/bind/zones# DHCP.vusna.com

Ab jasy previous lab mai host IP address k zariya access kiya tha ab on host IP address ko DNS assign kar os k through access karny ki Koshish karty hy

4. To modify the configuration of it01
    - root@web-server:/etc/apache2/sites-available# vi it601.conf

change the following settings

DirectoryIndex index.php index.html

ServerAdmin webmaster@vusna.com

ServerName web.vusna.com

ServerAlias www.web.vusna.com

Ab systemctl ki command sy restart kar k osska status check Kary gy

mahjabeen97669@gmail.com

5. To check that that the system is accessible by DNS or not
    - root@web_server:/etc/apache2/sites-available# curl web.vusna.com/it601/

LAB 09

single web server py multiple web services ko configure karny ko daikhy gy multi services use different domain names path k zarorat ni hoti sab mai

we have two Major task for this

1- configure multisites on web server

2- Add new FQDN in DNS server

1. To check the configuration of web server and available sites fr os ki files ko daikhy gy
    - root@web-server:/# cd/etc/apache2/

2. To go to the main configuration of it601

    - root@web_server:/etc/apache2/sites-available# vi it601.conf

3. Ab ider server ki Kuch setting ko change kry gy

DirectoryIndex index.php index.html

ServerAdmin webmaster@it601.vusna.com

ServerName it601.vusna.com

ServerAlias www.it601.vusna.com

document root /var/www/it601

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System and Network Administration
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[mahjabeen97869@gmail.com]

mahjabeen97869@gmail.com

same for it601p bas it601 ki jgah it601p ho jyy ga

4. To disable these sites
   - root@web_server:/etc/apache2/sites-available# a2dissites it601.conf

5. To enable these sites
   - root@web_server:/etc/apache2/sites-available# a2ensites it601.conf

6. To reload apache service
   - root@web_server:/etc/apache2/sites-available# systemctl apache2

7. To check the status of apache2
   - root@web_server:/etc/apache2/sites-available# systemctl status apache2

8. To change the DNS configuration first we check the existing configuration and local ko Check Kary gy
   - root@dns-server:/etc/bind# vi named.conf.options
   - root@dns-server:/etc/bind# vi named.conf.local

9. To go to zones
   - root@dns-server:/etc/bind# cd zones

here we have two files vusna.com and vusna.com.rev

10. To go to vusna .com and change its configuration
    - root@dns-server:/etc/bind/zones# vi vusna.com

VUSNA.COM Zone Configuration $TTL 604800

@ IN SOA vusna.com. root.vusna.com. (3; 604800; 86400; 2419200; 604800)

;

@ IN NS bindserver.vusna.com.

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bindserver IN A 192.168.19.2

;web IN A 192.168.19.3

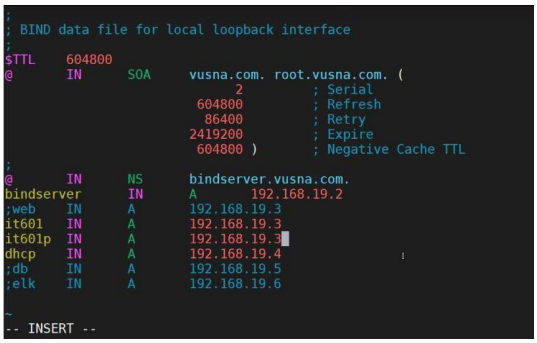
It601 IN A 192.168.19.3

It601p IN A 192.168.19.3

dhcp IN A 192.168.19.4

;db IN A 192.168.19.5

;elk IN A 192.168.19.6



colon Laga k ; comment out kar dain gy

or it601 and it601p k record ko b add Kary gy

11. To go to vusna .com.rev and change its configuration
    - root@dns-server:/etc/bind/zones# vi vusna.com

VUSNA.COM Reverse Zone Configuration $TTL 604800 @ IN SOA vusna.com. root.vusna.com. (2 ; 604800 ; 86400 ; 2419200 ; 604800)

;

@ IN NS bindserver.vusna.com.

bindserver IN A 192.168.19.2

2 IN PTR bindserver.vusna.com

;3 IN PTR web.vusna.com

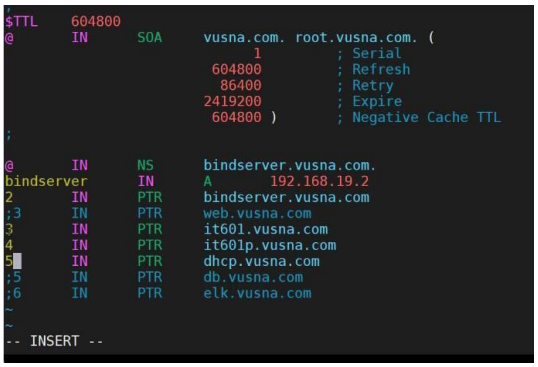
3 IN PTR it601.vusna.com

4 IN PTR it601p.vusna.com

5 IN PTR dhcp.vusna.com

;5 IN PTR db.vusna.com

;6 IN PTR elk.vusna.com



12. To restart the bind server and check it's status
    - root@dns-server:/etc/bind/zones# systemctl restart bind9
    - root@dns-server:/etc/bind/zones# systemctl status bind9

mahjabeen97869@gmail.com

13. To ping the host
    - root@dns-server:/etc/bind/zones# ping it601.vusna.com
    - root@dns-server:/etc/bind/zones# ping it601p.vusna.com

14. To install curl utility
    - root@dns-server:/etc/bind/zones# apt install curl

15. To check kaya ham different web services ki information retrieve kar sakty hy ya nai
    - root@dns-server:/etc/bind/zones# curl it601.vusna.com
    - root@dns-server:/etc/bind/zones# curl it601p.vusna.com

- Tasks

Configure Web Servers for two sites with FQDN

1 – it601.vusna.com

2 – it601p.vusna.com

- Steps

1- Create two conf files in /etc/apache2/

1- it601.conf
2- it601p.conf

2 – Create two content folders for it601 and it601p.

2 - Update ServerAdmin, ServerName and ServerAlias and Document root

3 – Reload Apache Server

4 – add A and PTR records for both services to web server

1 – comment out the existing web records
2 – Add A record for it601.vusna.com
3 – Add A record for it601.vusna.com
4 – Add PTR record for it601.vusna.com
5 – Add PTR record for IT601p.vusna.com

mahjabeen97869@gmail.com

5– Reload DNS Server

6 - Test the new services with cURL

LAB 10

protecting and configuring web services behaviour with .htaacces

sab sy pahly ham authentication ko daikhy gy

To enable such type of authentication we need square path first ta k ham apny passwords and users ki info ko save kar saky

1. To create web passes directory
   - root@web-server:/#mkdir web_passes

2. To create directory in web_passes
   - root@web-server:~/web_passes# htapasswd -c it601_user1 user1

then password ko authentication kr dain gy

3. To enable authentication on it601 pahly check kry gy koi pahly sy hi htaccess file to ni hy ager hy to osko remove kar dy gy
   - root@web-server:/var/www/it601# rm .rf . htacces

4. To create file htaccess than edit it
   - root@web-server:/var/www/it601# vi htaccess

5. To check the file location
   - Authuserfile /root/web_passes/it601_user1

6. group authentication ko null kar dy gy

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- AuthgroupFile /dev/null

7. user ko Jo prompt msg display karwana chaty osko ider likhy gy

Authname "The content you are accessing is protected!, please Enter your user name and password"

Authtype Basic

require valid-user

Now our configuration is complete Wq k sath isko write meansave kar dy gy

8. To reload the system
   - root@web-server:/var/www/it601# systemctl reload apache2

9. To change the permission of htaccess tak k probably executable and Accessable ho
   - root@web-server:/var/www/it601# chmod 644 htaccess

Ab ham Kuch configuration define Kary gy k ager user unprotactive way ay access ki try Kary to ham osko automatically https ki traf divert kar dain

10. To pahly htaccess ki file create Kary gy
    - root@web-server:/var/www/it601#vi htaccess

For redirection we use rewrite engine

- now we can describe some conditions that are given below

RewriteEngine on (on the engine)

RewriteCond %{HTTPS } !=on

(it is used to write condition mtlb ager http on ni hy to ham osko redirect kr dy gy)

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RewriteRule ^/(.*) https://{SERVER_NAME} /$1[R,L]

(yha ham ik rule define kr dy gy k beginning of url it could be anything isko ham https py divert kar dy gy

11. To save and change its permission
    - root@web-server:/var/www/it601#chmod -R 777 . htaccess

then isko restart kar ly gy

is trah ham or b boht sy changes kar sakty koi restrictions lga sakty k Kon hmari web ko access kar Sakta Kon nai to fr ham visitors ko block kar sakty ta k web services k oper load km ho

Configurations with htaccess

.htaccess

➢ The .htaccess file in Apache is a tool that allows configurations at the directory and subdirectory level.

➢ Using .htaccess enables you to configure website permissions without altering server configuration files.

➢ **Enabling .htaccess**

▪ sudo vi /var/www/my_website.com/.htaccess

▪ sudo vi /user/safe_location/.htpasswd

➢ **Common Uses**

▪ Manage IP Addresses

▪ Block Visitors by Referrer

▪ Redirect Traffic

▪ Set a 404 Page

Enable authentication

1 – Create password with htpasswd on secure location

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- htpasswd –c <secure_location> <password_filename> <user_id>

2 – Create .htaccess file in the folder to be protected

- vi .htaccess

3 – Write the following lines in above file

AuthUserFile /user/safe_location/.htpasswd

- AuthGroupFile /dev/null
- AuthName "Please Enter Password"
- AuthType Basic
- Require valid-user

4 - Reload the apache2

Allow. Deny IP Addresses

1 – Add the following lines to the htaccess file in the folder where restrictions are to be applied.

- order deny, allow
- deny from 192.168.0.54
- allow from 192.168.0

Set a 404 Page

1 – Add the following lines to the htaccess file.

- ErrorDocument 404 /404.html

➢Redirect Traffic

1 – Add the following lines to the htaccess file in the folder where redirection is to be applied.

- Redirect301/Other_Website.com/index.html/My_Website.com/index.html

Block Visitors by Referrer

1 – Add the following lines to the htaccess file in the folder where restrictions are to be applied

- RewriteEngine on
- # Options +FollowSymlinks
- RewriteCond %{HTTP_REFERER} blockeddomain\.com [NC]

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- RewriteRule .* - [F]

LAB 11

Linux server py kerberos ko install kry gy

fr oski setting kry gy pahly waly sab server ko root user sy access kar ry thy sab server ko real life ham asy sab mai ni kab sakty is k is k Liya ham IAS solution bnay gy jis sy different user ko access ho gi server py

trah hmary pass ik centralized location mil jyy gi or isi k zariya hm kdc server portal sy sab ko mange Kary gy

is maqsad k Liya ham ny new vm B'nai hy then

1) To connect the kerberos with it's console
    - it601p@labserver : ~$ sudo docker -it kerberos_server bash
    - [sudo] password for it601p: it601p

now we are login on this server

2) To to check it's IP address
    - root@kerberos:/# ifconfig

3) To update the repositories
    - root@kerberos:/# apt update

4) To intall two packages that we mentioned
    - root@kerberos:/# apt install krb5-kdc krb5-admin-server
    - then iska realm dy gy :vusna.com
    - then iska kerberos server dy : kerberos.vusna.com
    - then administrative server dy gy : kerberos.vusna.com

Now package installation is done

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Now we have to set the password for realm local database

5) To set password for realm local database
   - root@kerberos:/# krb5-newrealm
   - then master key ki jgah password dy dain gy

6) Ager installation k time ham ny koi koi mistake ki ho to ham osko is command sy again configure kr sakty
   - root@kerberos:/# sudo dpkg-reconfigure krb5-kdc

7) To add first principle
   - kadmin.local:addprinc sna
   - no policy ka option ayy ga

then iska password set kar dy gy. so new principle create ho jyy ga

to exit we can type quit on k admin local

8) To add admin principal
   - root@kerberos:/# kadmin local
   - k admin local : addprinc sna/admin

password same oper wala rakhy gy Jo pahly diya

thek quit kar dy gy

9) To add the entry in etc
   - root@kerberos:/# vu /etc/krb5kdc/kadm.acl

10) To set the principal
   - sna/admin@vusna.com * (*) is used for all permission

mahjabeen97869@gmail.com

ab is ki wq kar k save kar dy gy

11) To restart the service

- root@kerberos:/# systemctl restart krb5-admin-server.service

12) To test the principal

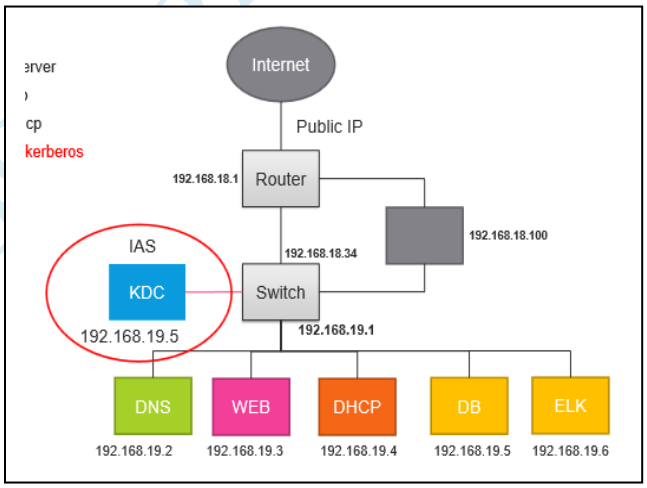
- root@kerberos:/# kinit sna/admin

➢To check more information

root@kerberos:/# klist

Kerberos IAS

1. Local Networks : 192.168.19.0/24
2. DNS SERVER: 192.168.19.2 , bindserver
3. WEB_SERVER : 192.168.19.3 , web
4. DHCP_SERVER : 192.168.19.4 , dhcp
5. kerberos_SERVER : 192.168.19.5 , kerberos
6. ELK_SERVER : 192.168.19.6 , elk
7. DOMAIN : vusna.com



Tasks

- Realm: vusna.com
- Primary KDC: kerberos.vusna.com
- User principal: sna
- Admin principal: sna/admin

Prerequisites

manjabeen97866@gmail.com

1- Setup the new server machine

2- assign IP addressing, SSH and Console Access

3- Set hostname

KDC Installation

1 - Install the Kerberos Packages

- **krb5-kdc**
- **krb5-admin-server**
- **sudo apt install krb5-kdc krb5-admin-server**

2 - Create the new realm with the kdb5_newrealm utility

- **sudo krb5_newrealm**

3 - you can reconfigure Kerberos from scratch, if need to change the realm name. Following commands can be used for this purpose.

- **sudo dpkg-reconfigure krb5-kdc**

KDC Configuration

4 - Create the first principal.

- sudo kadmin.local
- Authenticating as principal root/admin@vusna.com with password.
- kadmin.local: addprinc sna
- WARNING: no policy specified for sna@vusna.com; defaulting to no policy
- Enter password for principal “sna@vusna.com":
- Re-enter password for principal " sna@vusna.com “:
- Principal “sna@ vusna.com" created.
- kadmin.local: quit

Create Admin Principal

5 - Create an admin principal and assign permissions.

- $ sudo kadmin.local
- Authenticating as principal root/admin@vusna.com with password.

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- kadmin.local: addprinc sna/admin
- WARNING: no policy specified for sna/admin@vusna.com; defaulting to no policy
- Enter password for principal " sna/admin@vusna.com ":
- Re-enter password for principal " sna/admin@vusna.com ":
- Principal "sna/admin@vusna.com" created.
- kadmin.local: quit

Create Admin Principal

6 - Set admin principal with appropriate Access Control List (ACL) permissions. The permissions

are configured in the /etc/krb5kdc/kadm5.acl file:

- **sna/admin@vusna.com ***

The above will grant all privileges to any admin instance of a principal. See the kadm5.acl manpage for

details.

7 - Restart the krb5-admin-server for the new ACL to take affect:

- **sudo systemctl restart krb5-admin-server.service**

8 - The new user principal can be tested using the kinit utility:

- $ kinit sna/admin
- Password for sna/admin@vusna.com:

Verifying and Completing Kerberos

9 - Verify information about the Ticket Granting Ticket (TGT):

- $ klist
- Ticket cache: FILE:/tmp/krb5cc_1000
- Default principal: sna/admin@vusna.com
- Valid starting Expires Service principal
- 04/03/20 19:16:57 04/04/20 05:16:57 krbtgt/vusna.com@vusna.com
- renew until 04/04/20 19:16:55

➢ Your new Kerberos Realm is now ready to authenticate clients.

mahjabeen97869@gmail.com

LAB 12

In today lab ham kerberos ki installation k sath ldap ko integrate Kary gy

ldap is a light weight directory access protocol ya basically authentication or directory ki services provide karta hy and it is based on x500 protocol Jo information ko tree ki form mai store karta hy is sy Hamy information ko search karny mai asani rahti or ham is ko easily authenticate b kar sakty or user ko allocate b kar sakty hy

We use it as database for kerberos and iska php boy account manager ka portal hy wo b install Kary gy ta k ham command line k bjyy gui base tool sy user ko manage kar saky

1) To update repositories
   - root@kerberos:/# apt update

2) To install ldap utilities
   - root@kerberos:/# apt install slapd ldap-utils

3) To configure the package that are not automatically configure
   - root@kerberos:/# sudo dpkg-reconfigure slapd

Next wo ham sy domain name pochy ga that is vusna.com

organization name = cds

Admin password= abcd

4) To configure the OpenLDAP libraries with certain defaults

we go to

- root@kerberos:/# /etc/ldap/ldap.conf

BASE dc=vusna,dc=com

manjabeen97869@gmail.com

URI ldap://ldap.vusna.com

5) To check k Jo ldap ka system ham ny install kiya hy wo Kam kar b rha hy ya ni
   - root@kerberos:/etc /ldap# ldapsearch -Q -LLL -Y EXTERNAL -H ldapi:/// -b cn=config dn

6) To search the specific information about vusna
   - root@kerberos:/etc /ldap# ldapsearch -x -LLL -H ldap:/// -b dc=vusna,dc=com dn

7) To create directory information tree jis Mai people ho gy or group hogy is maqsad k Liya Hamy ldif ki file create Karni ho gi
   - root@kerberos:/etc /ldap# ldap_data.ldif

Edit mode py ja k ham APNA data copy kar k wha paste Kary gy

dn: ou=People,dc=vusna,dc=com

objectClass: organizationalUnit

ou: People

dn: ou=Groups,dc= vusna,dc=com

objectClass: organizationalUnit

ou: Groups

dn: cn=csd,ou=Groups,dc=

vusna,dc=com

objectClass: posixGroup

cn: csd

gidNumber: 5000

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dn: uid=arif,ou=People,dc=vusna,dc=com

objectClass: inetOrgPerson

objectClass: posixAccount

objectClass: shadowAccount

uid: arif

sn: husen

givenName: Rashid

cn: Arif Rashid

displayName: Arif H Rashid

uidNumber: 10000

gidNumber: 5000

userPassword: {CRYPT}x

gecos: Arif Rashid

loginShell: /bin/bash

homeDirectory: /home/arif

8) To add the ldif data

- root@kerberos:/etc /ldap# ldapadd -x -D cn=admin,dc=vusna.com,dc=com -W -f ldap_data.ldif

Enter LDAP Password: ****

Now ldap installation is complete

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Ab ham isi server py account manager ko install kry gy

9) To install the packages to install account manager

- root@kerberos:/etc /ldap# apt -y install apache2 php php-cgi libapache2-mod-php php-mbstring php-common php-pear

10) To enable php-cgi PHP extension.

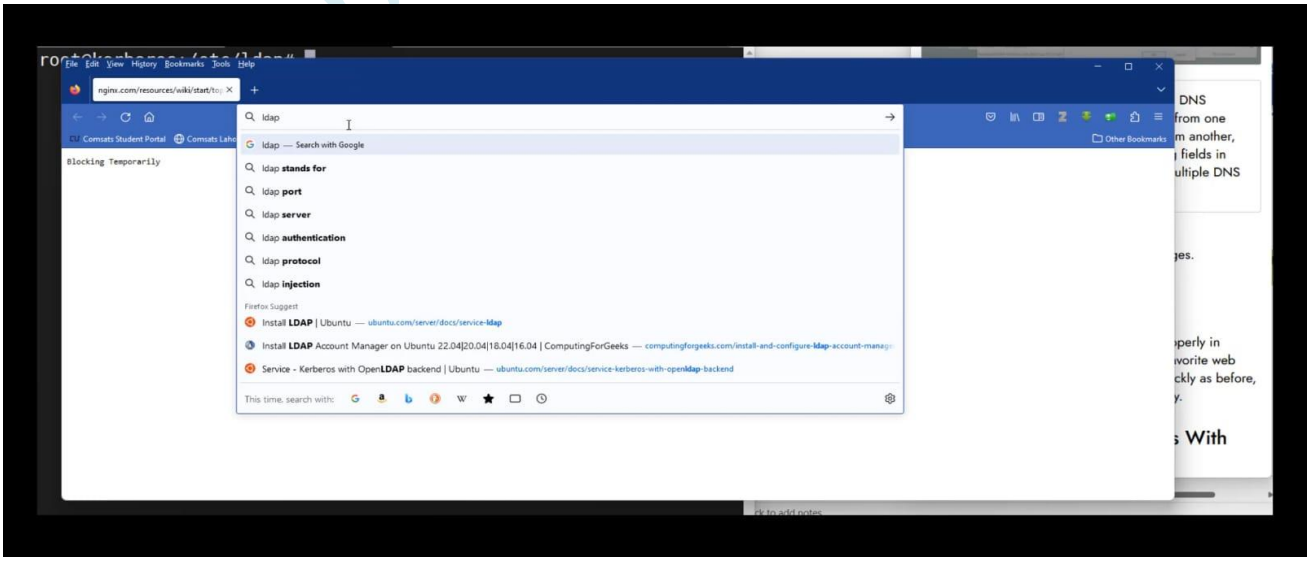
- sudo a2enconf php*-cgi
- sudo systemctl reload apache2

To install apache2 LDAP account manager

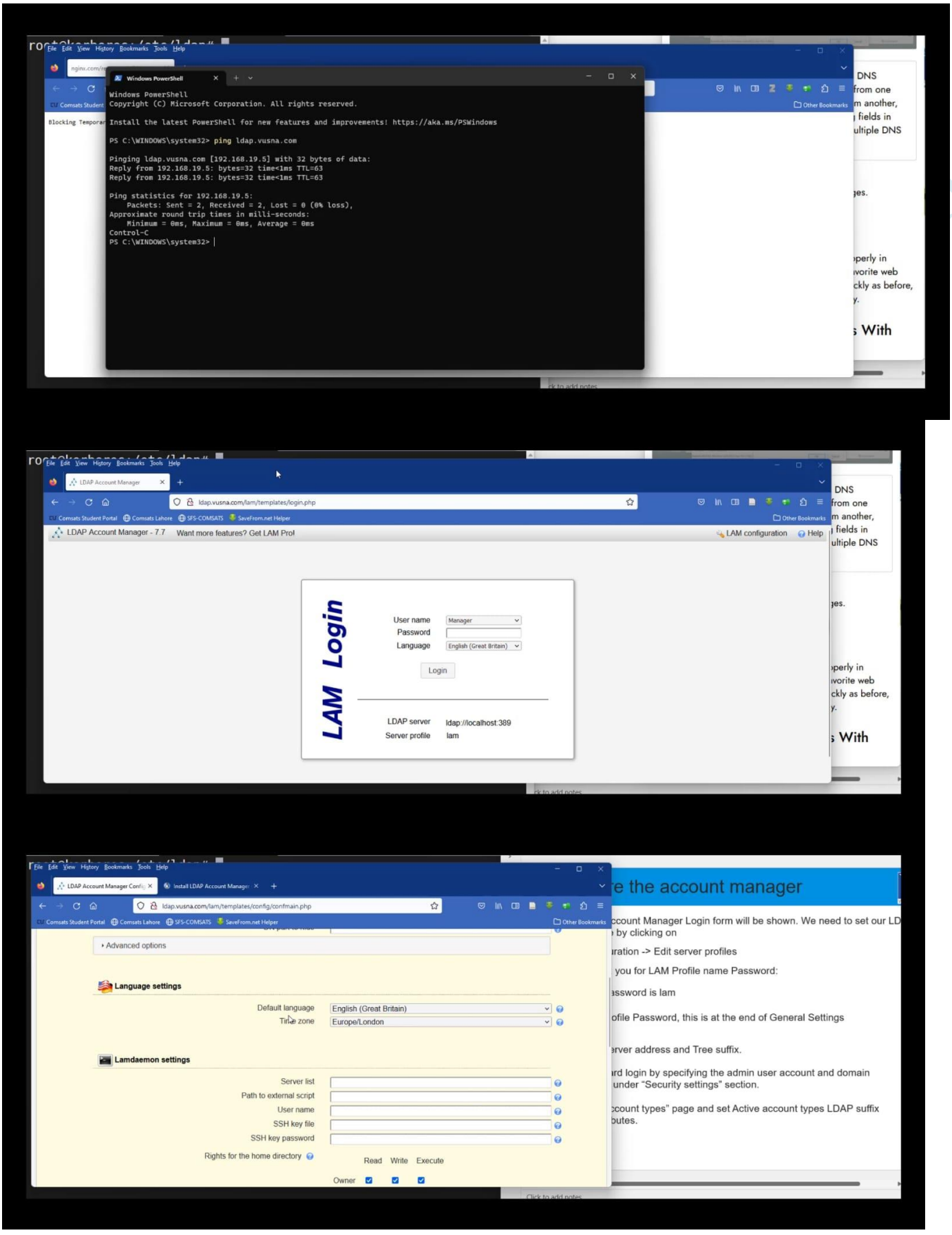
- root@kerberos:/etc /ldap# apt -y install ldap-account-manager

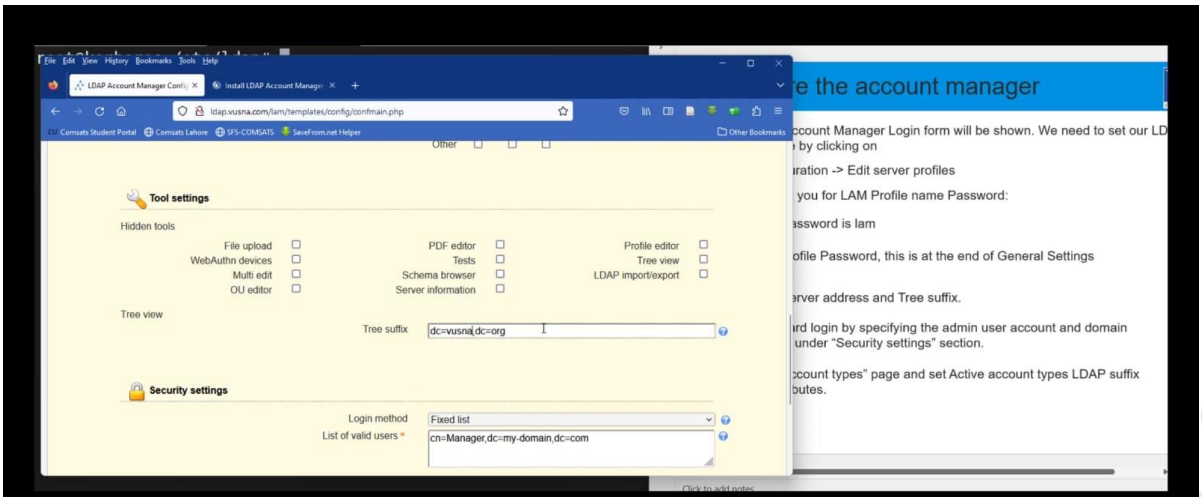
Now our LDAP account manager is installed

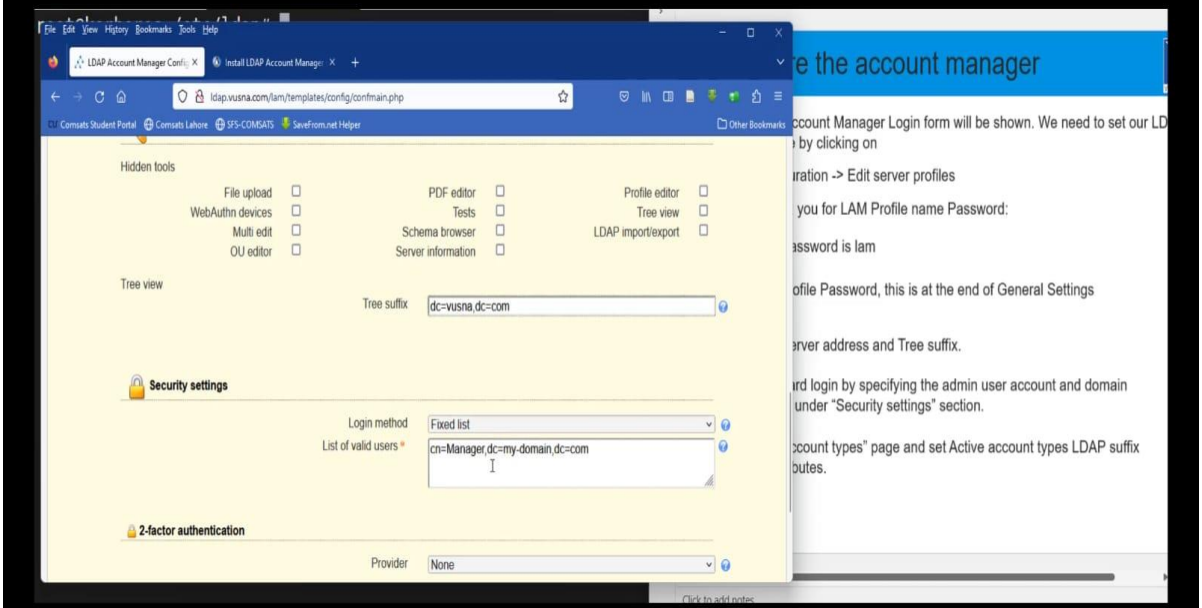
Ab ham isko apny window k host sy access karny ki try karty hy

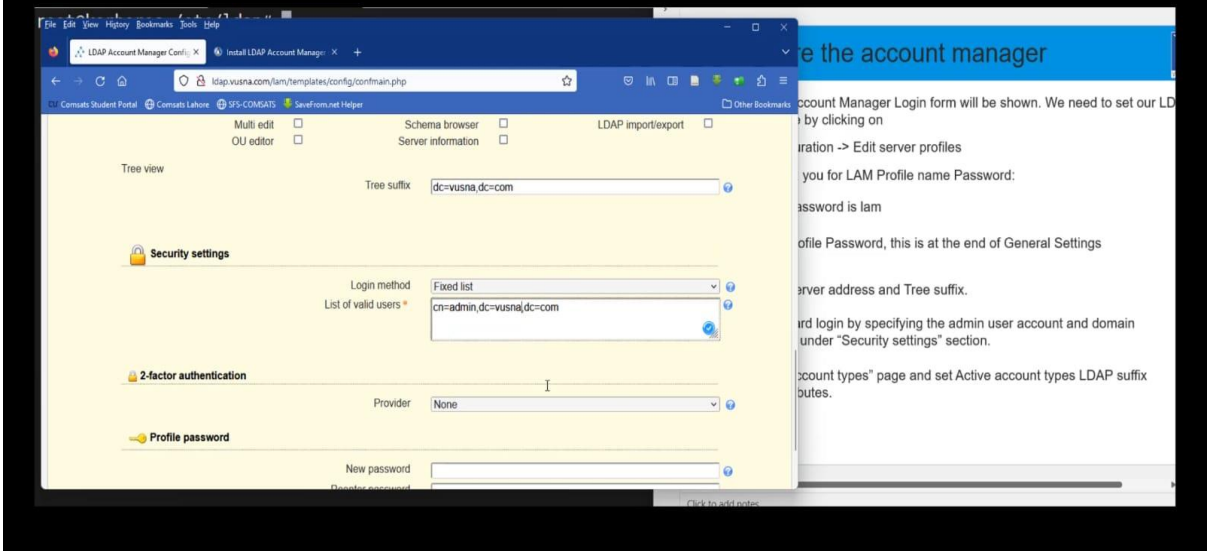


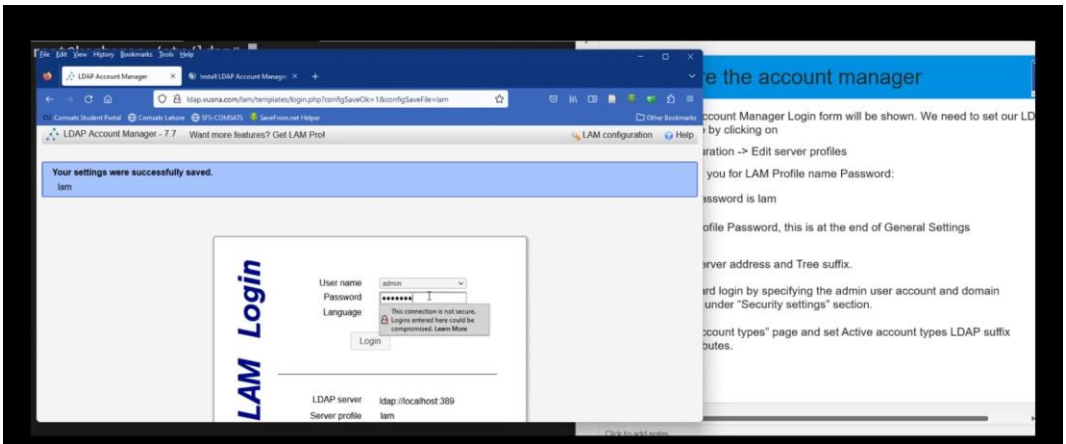
been97869@gmail.com

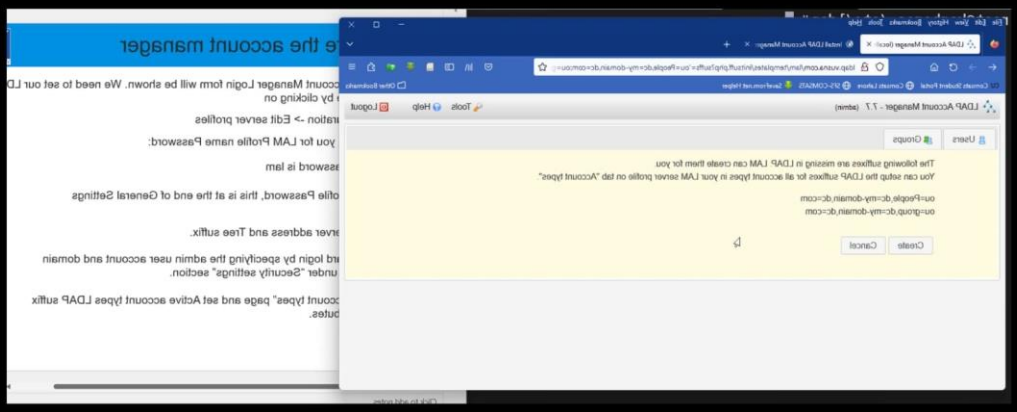

Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows
PS C:\WINDOWS\system32> ping ldap.vusna.com
Pinging ldap.vusna.com [192.168.19.5] with 32 bytes of data:
Reply from 192.168.19.5: bytes=32 time<1ms TTL=63
Reply from 192.168.19.5: bytes=32 time<1ms TTL=63
Ping statistics for 192.168.19.5:
Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
Control-C
PS C:\WINDOWS\system32>
LDAP Account Manager
ldap.vusna.com/lam/templates/login.php
LDAP Account Manager - 7.7
Want more features? Get LAM Pro!
LAM configuration
Help
LAM Login
User name
Manager
Password
Language
English (Great Britain)
Login
LDAP server
ldap://localhost:389
Server profile
lam
Advanced options
Language settings
Default language
English (Great Britain)
Time zone
Europe/London
Lamdaemon settings
Server list
Path to external script
User name
SSH key file
SSH key password
Rights for the home directory
Read
Write
Execute
Owner

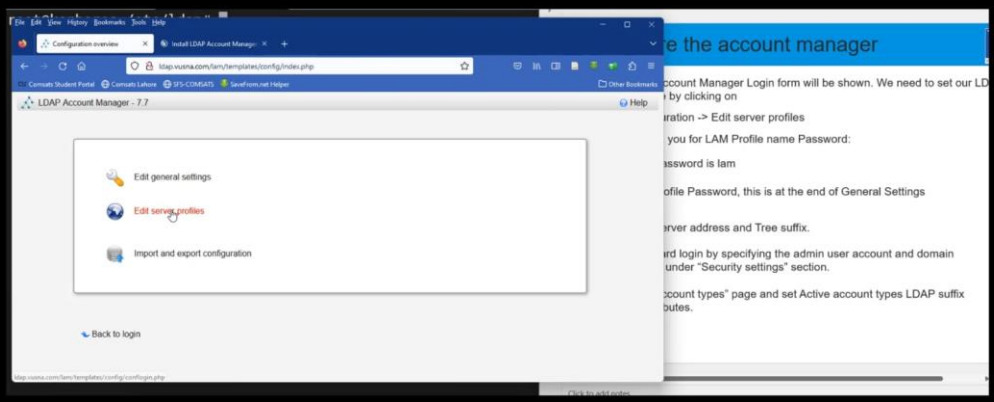

ldap.vusna.com/lam/templates/config/confmain.php
Tool settings
Hidden tools
File upload
WebAuthn devices
Multi edit
OU editor
PDF editor
Tests
Schema browser
Server information
Profile editor
Tree view
LDAP import/export
Tree view
Tree suffix
dc=vusna,dc=org
Security settings
Login method
Fixed list
List of valid users *
cn=Manager,dc=my-domain,dc=com
the account manager

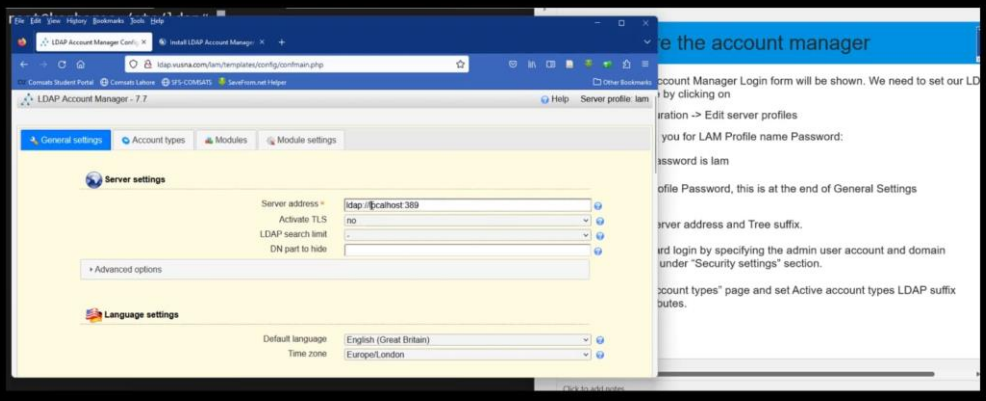

ldap.vusna.com/lam/templates/config/confmain.php
Hidden tools
File upload
WebAuthn devices
Multi edit
OU editor
PDF editor
Tests
Schema browser
Server information
Profile editor
Tree view
LDAP import/export
Tree view
Tree suffix
dc=vusna,dc=com
Security settings
Login method
Fixed list
List of valid users *
cn=Manager,dc=my-domain,dc=com
2-factor authentication
Provider
None
the account manager

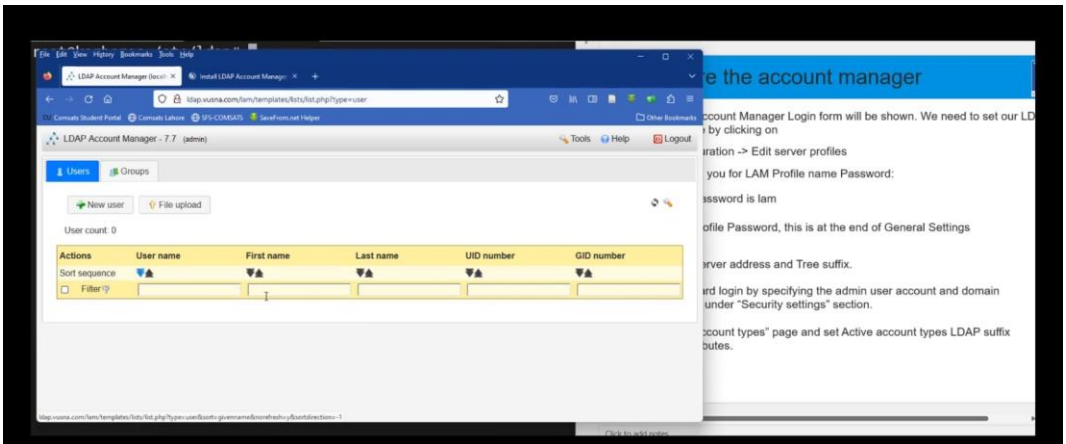

ldap.vusna.com/lam/templates/config/confmain.php
Multi edit
OU editor
Schema browser
Server information
LDAP import/export
Tree view
Tree suffix
dc=vusna,dc=com
Security settings
Login method
Fixed list
List of valid users *
cn=admin,dc=vusna,dc=com
2-factor authentication
Provider
None
Profile password
New password
the account manager

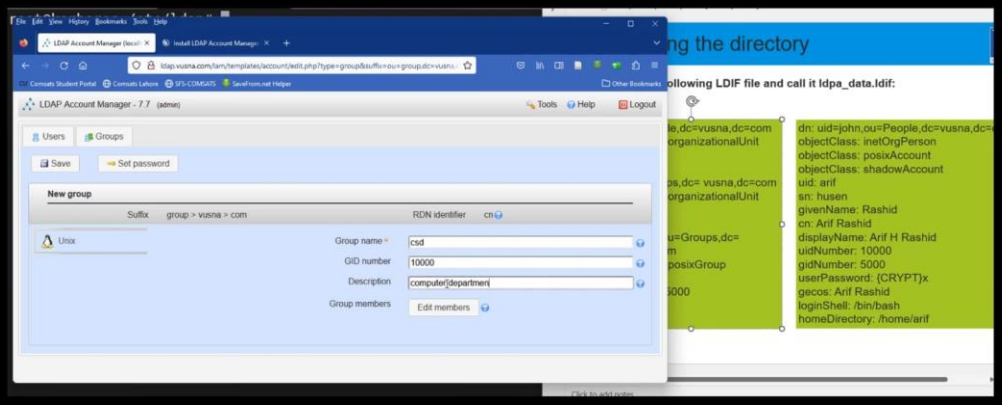

LDAP Account Manager - 7.7
Your settings were successfully saved.
lam
LAM Login
User name
admin
Password
Language
Login
LDAP server
ldap://localhost:389
Server profile
lam

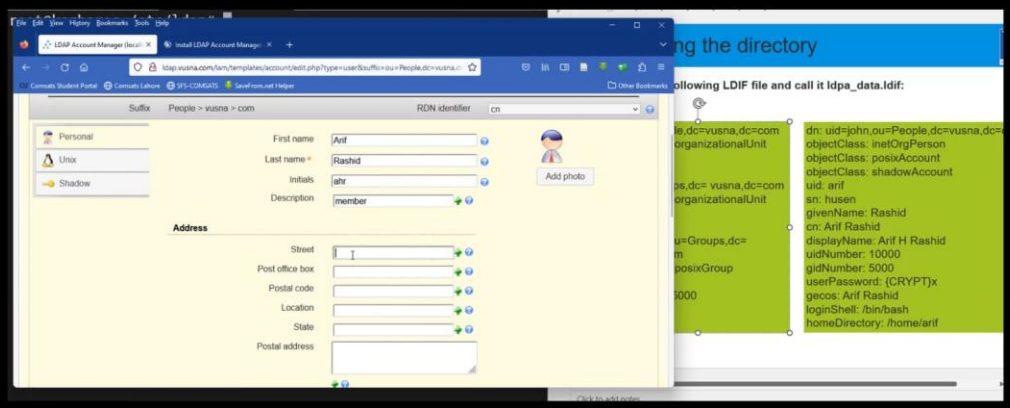


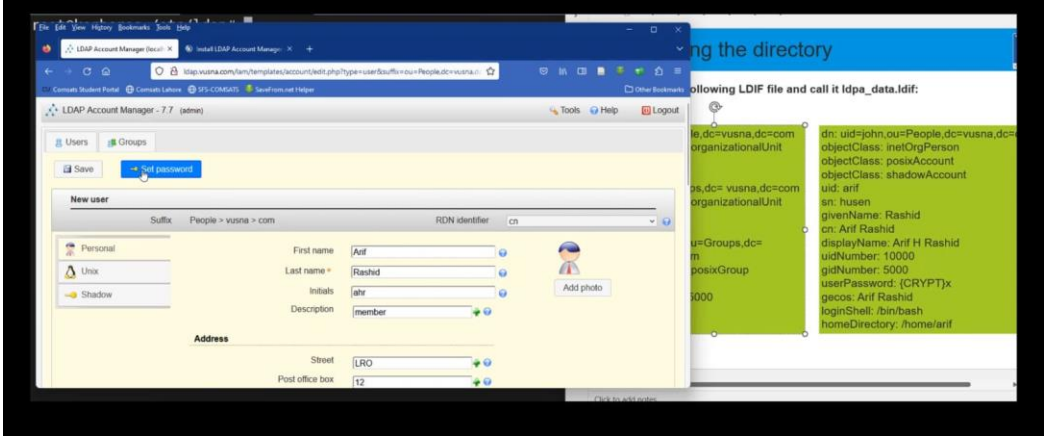

LDAP Account Manager - 7.7
Edit general settings
Edit server profiles
Import and export configuration
Back to login

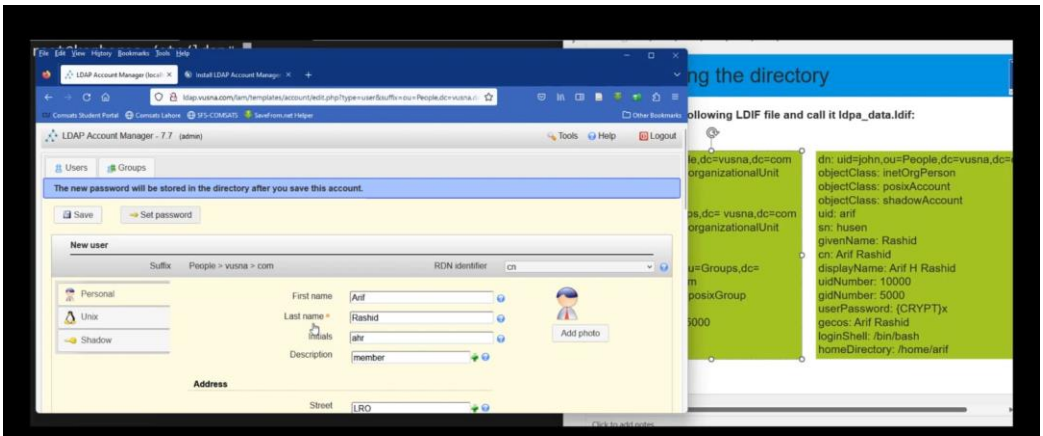

LDAP Account Manager - 7.7
Help
Server profile: lam
General settings
Account types
Modules
Module settings
Server settings
Server address
ldap://localhost:389
Activate TLS
no
LDAP search limit
DN part to hide
Advanced options
Language settings
Default language
English (Great Britain)
Time zone
Europe/London

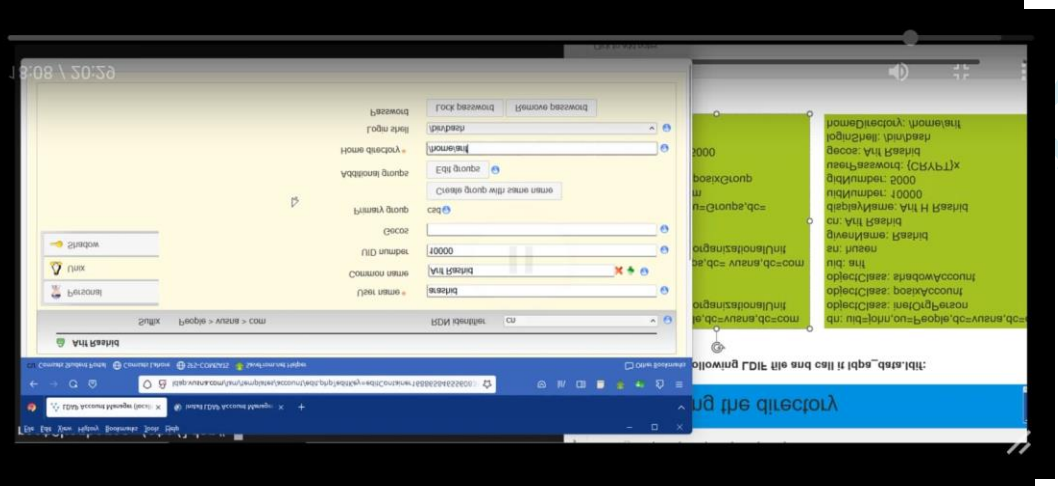

LDAP Account Manager - 7.7 (admin)
Tools
Help
Logout
Users
Groups
New user
File upload
User count: 0
Actions
User name
First name
Last name
UID number
GID number
Sort sequence
Filter
e the account manager
count Manager Login form will be shown. We need to set our LD
by clicking on
iration -> Edit server profiles
you for LAM Profile name Password:
ssword is lam
ofile Password, this is at the end of General Settings
rver address and Tree suffix.
rd login by specifying the admin user account and domain
under "Security settings" section.
count types" page and set Active account types LDAP suffix
butes.

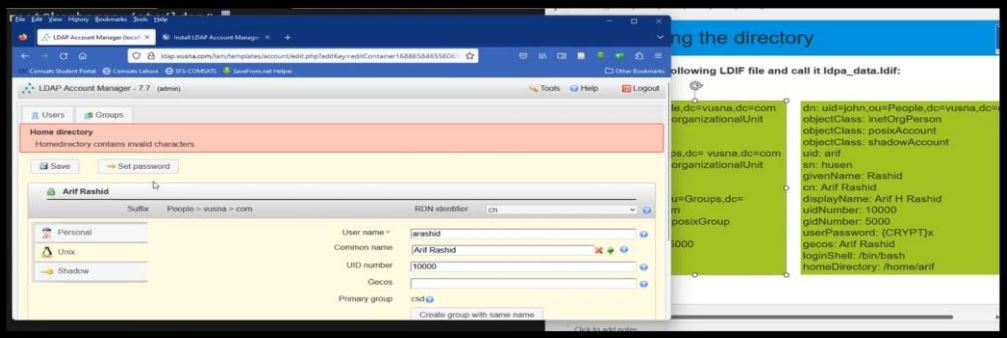

LDAP Account Manager - 7.7 (admin)
Tools
Help
Logout
Users
Groups
Save
Set password
New group
Suffix
group > vusna > com
RDN identifier
cn
Unix
Group name *
csd
GID number
10000
Description
computer|departmen
Group members
Edit members
ng the directory
llowing LDIF file and call it ldpa_data.ldif:
dn: uid=john,ou=People,dc=vusna,dc=
objectClass: inetOrgPerson
objectClass: posixAccount
objectClass: shadowAccount
uid: arif
sn: husen
givenName: Rashid
cn: Arif Rashid
displayName: Arif H Rashid
uidNumber: 10000
gidNumber: 5000
userPassword: {CRYPT}x
gecos: Arif Rashid
loginShell: /bin/bash
homeDirectory: /home/arif

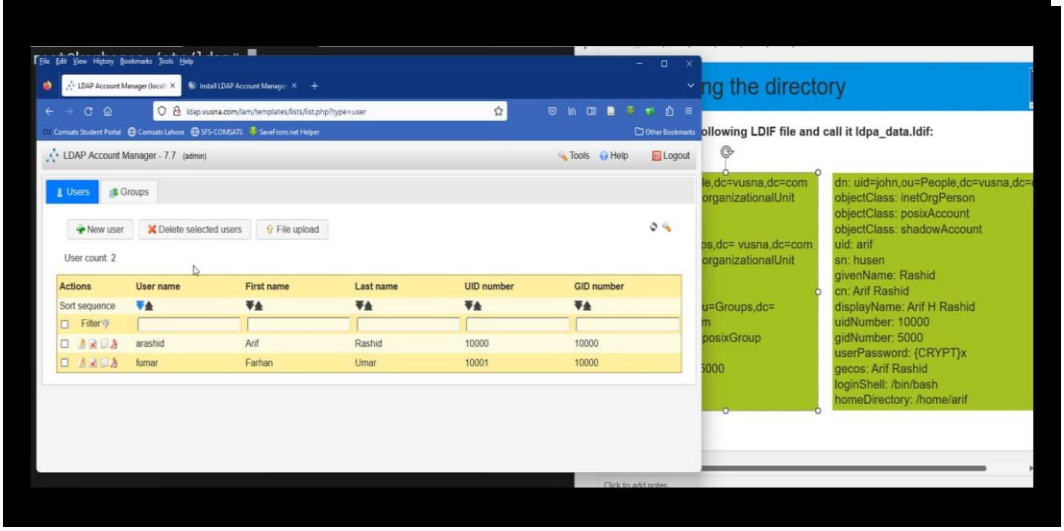

Suffix
People > vusna > com
RDN identifier
cn
Personal
Unix
Shadow
First name
Arif
Last name *
Rashid
Initials
ahr
Description
member
Add photo
Address
Street
Post office box
Postal code
Location
State
Postal address
ng the directory
llowing LDIF file and call it ldpa_data.ldif:
dn: uid=john,ou=People,dc=vusna,dc=
objectClass: inetOrgPerson
objectClass: posixAccount
objectClass: shadowAccount
uid: arif
sn: husen
givenName: Rashid
cn: Arif Rashid
displayName: Arif H Rashid
uidNumber: 10000
gidNumber: 5000
userPassword: {CRYPT}x
gecos: Arif Rashid
loginShell: /bin/bash
homeDirectory: /home/arif


LDAP Account Manager - 7.7 (admin)
Tools
Help
Logout
Users
Groups
Save
Set password
New user
Suffix
People > vusna > com
RDN identifier
cn
Personal
Unix
Shadow
First name
Arif
Last name *
Rashid
Initials
ahr
Description
member
Add photo
Address
Street
LRO
Post office box
12
ng the directory
llowing LDIF file and call it ldpa_data.ldif:
dn: uid=john,ou=People,dc=vusna,dc=
objectClass: inetOrgPerson
objectClass: posixAccount
objectClass: shadowAccount
uid: arif
sn: husen
givenName: Rashid
cn: Arif Rashid
displayName: Arif H Rashid
uidNumber: 10000
gidNumber: 5000
userPassword: {CRYPT}x
gecos: Arif Rashid
loginShell: /bin/bash
homeDirectory: /home/arif


LDAP Account Manager - 7.7
The new password will be stored in the directory after you save this account.
New user
First name Arif
Last name Rashid
Initials ahr
Description member
Address
Street LRO
ng the directory
following LDIF file and call it ldpa_data.ldif:
dn: uid=john,ou=People,dc=vusna,dc=
objectClass: inetOrgPerson
objectClass: posixAccount
objectClass: shadowAccount
uid: arif
sn: husen
givenName: Rashid
cn: Arif Rashid
displayName: Arif H Rashid
uidNumber: 10000
gidNumber: 5000
userPassword: {CRYPT}x
gecos: Arif Rashid
loginShell: /bin/bash
homeDirectory: /home/arif




LDAP Account Manager - 7.7
Home directory
Homedirectory contains invalid characters.
Arif Rashid
User name arashid
Common name Arif Rashid
UID number 10000
Gecos
Primary group csd
Create group with same name
ng the directory
following LDIF file and call it ldpa_data.ldif:
dn: uid=john,ou=People,dc=vusna,dc=
objectClass: inetOrgPerson
objectClass: posixAccount
objectClass: shadowAccount
uid: arif
sn: husen
givenName: Rashid
cn: Arif Rashid
displayName: Arif H Rashid
uidNumber: 10000
gidNumber: 5000
userPassword: {CRYPT}x
gecos: Arif Rashid
loginShell: /bin/bash
homeDirectory: /home/arif


LDAP Account Manager - 7.7
Users
Groups
New user
Delete selected users
File upload
User count: 2
Actions
User name
First name
Last name
UID number
GID number
Sort sequence
Filter
arashid Arif Rashid 10000 10000
fumar Farhan Umar 10001 10000
ng the directory
following LDIF file and call it ldpa_data.ldif:
dn: uid=john,ou=People,dc=vusna,dc=
objectClass: inetOrgPerson
objectClass: posixAccount
objectClass: shadowAccount
uid: arif
sn: husen
givenName: Rashid
cn: Arif Rashid
displayName: Arif H Rashid
uidNumber: 10000
gidNumber: 5000
userPassword: {CRYPT}x
gecos: Arif Rashid
loginShell: /bin/bash
homeDirectory: /home/arif

window host py sari setting k bad jab ham ldap search ki command run Kary gy to ham ny Jo user enter kiyy hy waha wo b Hamy show hona shoro ho jayy gy

LAB 12 (PART 2)

Ab tak ham ny jitni b lab ki hy os mai jis jgah py encryption ya security ki zarorat thi to ham ny self sign digital certificates hi use kiyy thy in certificates ka issue ya hota hy k browser ya email client py wo isko recognize ni Karti or message ata hy k site trusted ni hy

Trusted sites k Liya Hamy local ca server setup Karna parta hy organization k Ander is k Kiya hamary pass open ssl ka package use karty hy apen ssl bunyadi Tor py ssl and tls ki implementation hy Jo open source Hain or internet py communication ko secure karny k Liya use hota hy

is k Ander Jo different qisam ki functionalities supported hoti Hain wo ya hy k ham rsa ,help me ya digital signature algorithm k key parameter create kar sakty Hain jasy k x509 certificates , certificate signing csr and crl etc is k ilawa is sy ham digest b calculate kar sakty hy encryption and decryption and mail ki functionalities b implement kar sakty Hain

To setup the ca we have to follow different steps

- firstly Install open ssl package for this purpose we can firstly update the repositories

But firstly ham check Kary gy k kaya open ssl ka package hmary system py install hy ya nai we use the command

- root@ca:/# which openssl

1) If open ssl is not install we can install it by apt command
   - root@ca:/# apt install openssl

create file and directories for ca certificate

2) To create directories
   - root@ca:/# mkdir -p /etc/ssl/vusnaCA {certs, private, crl, newcerts, csr}

3) To view our created directories

mahjabeen97869@gmail.com

- root@ca:/# cd /etc/ssl/

our directories exist ab isi k Ander hmari next directories b mojod Hain

root@ca: /etc/ssl #cd vusnaCA/

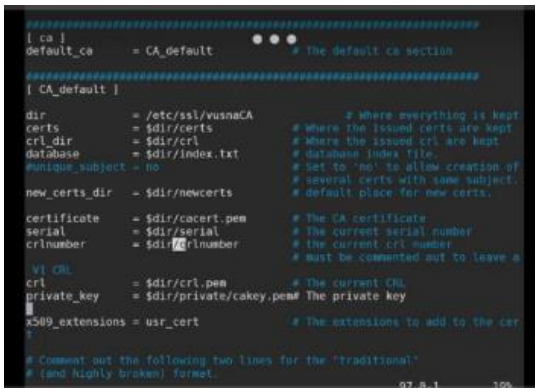
root@ca:/etc/ssl/vusnaCA#

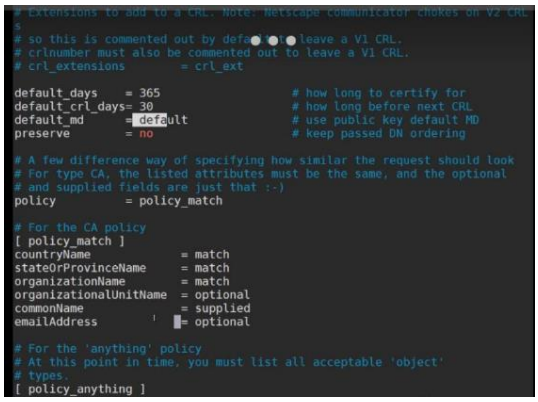
Open ssl k Ander Kuch changing Kary gy ham is k Liya ham pahly ssl ki original installation hoti hy os k ander ssl ki configuration Jo already mojod hoti hy isi file ko ham apny vusnaCA k
Ander copy kar lain gy

root@ca:/etc/ssl/vusnaCA# cp /etc/ssl/openssl.cnf /etc/ssl/vusnaCA

4) To open the configuration of open ssl with vi editor

root@ca:/etc/ssl/vusnaCA# vi /etc/ssl/vusnaCA/openssl.cnf





Yaha ham ny Pahla section CA ka change Karna hy

fr certificate ki life k about information add kry gy

is k bad requirements ka section change Kary gy

after validate all the information that are provided in screenshot
now go to the user certificate section

mahjabeen97869@gmail.com

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System and Network Administration
LAB SOLUTIONS
Made by Mah jabeen
[mahjabeen97869@gmail.com]

[usr_cert]

```
challengePassword_max           = 20

unstructuredName                = optional company name

[ usr_cert ]

# These extensions are added when 'ca' signs a request.

# This goes against PKIX guidelines but some CAs do it and some software
# requires this to avoid interpreting an end user certificate as a CA.

basicConstraints=CA:FALSE

# This is typical in keyUsage for a client certificate.
keyUsage = nonRepudiation, digitalSignature, keyEncipherment,dataEncipherm
ent

# PKIX recommendations harmless if included in all certificates.
subjectKeyIdentifier=hash
authorityKeyIdentifier=keyid,issuer

# This stuff is for subjectAltName and issuerAltname.
# Import the email address.
# subjectAltName=email:copy
# An alternative to produce certificates that aren't
# deprecated according to PKIX.
subjectAltName = @alt_names

[alt_names]
```

next is [alt_names]

```
# An alternative to produce certificates that aren't
# deprecated according to PKIX.
subjectAltName = @alt_names

[alt_names]

DNS.1 = *.vusna.com

# Copy subject details
# issuerAltName=issuer:copy

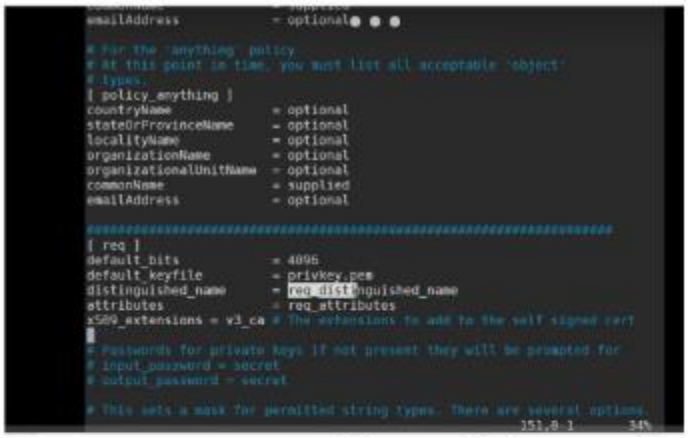
# This is required for TSA certificates.
# extendedKeyUsage = critical,timeStamping

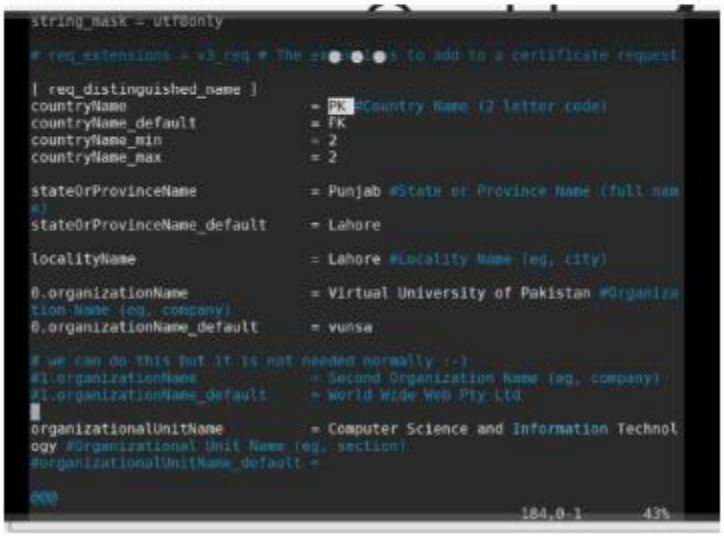
[ v3_req ]

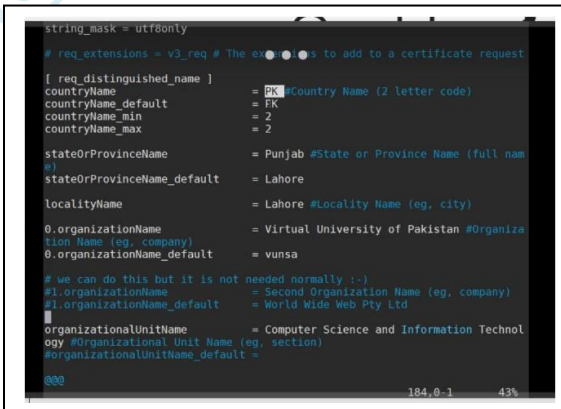
# Extensions to add to a certificate request
```

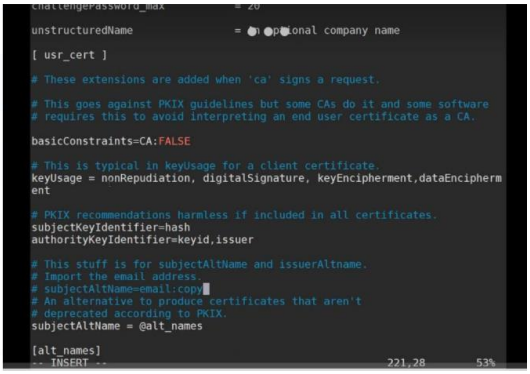
nonrepudiation mean k Jo b message

ham send karty hy user/sender osko denied ni kar Sakta because it is signed by his private key









ya hmari open ssl ki basic configuration thi is ko ham save kar ly gy

Now

5) To initialize our serial and index files
   - root@ca:/etc/ssl/vusnaCA# echo 01 > /etc/ssl/vusnaCA/serial

6) To create file
   - root@ca:/etc/ssl/vusnaCA# touch /etc/ssl/vusnaCA /index.txt

7) To generate private key for root
   - root@ca:/etc/ssl/vusnaCAa openssl genrsa -aes256 out /etc/ssl/vusnaCA/private/cakey.pem 4096

Enter PEM pass phrase:

Verifying Enter PEN pass phrase:

private key is generated

8) To create the root certificate

Verifying Enter PEM pass phrase:

root@ca:/etc/ssl/vusnaCA# openssl req new x509-sha256 \

> -config /etc/ssl/vusnaCA/openssl.cnf \

> -key /etc/ssl/vusnaCA/private/cakey.pem \

> -out /etc/ssl/vusnaCA/cacert.pem \

> -subj "/C=PK ST=Punjab\L=Lahore\0=vusna\CN=vusna.com" Enter pass phrase for /etc/ssl/vusnaCA/private/cakey.pem:

mahjabeen97869@gmail.com

9) To view the certificate that wee are generated now
   - root@ca:/etc/ssl/vusnaCA# openssl x509 -in /etc/ssl/vusnaCA /cacert.pem

10) To generate local ssl and tls signing request for this purpose we have to generate a key for our local domain vusna
   - root@ca:/etc/ssl/vusnaCA# openssl genrsa -out /etc/ssl/vusnaCA/private/vusna.com.key 4096

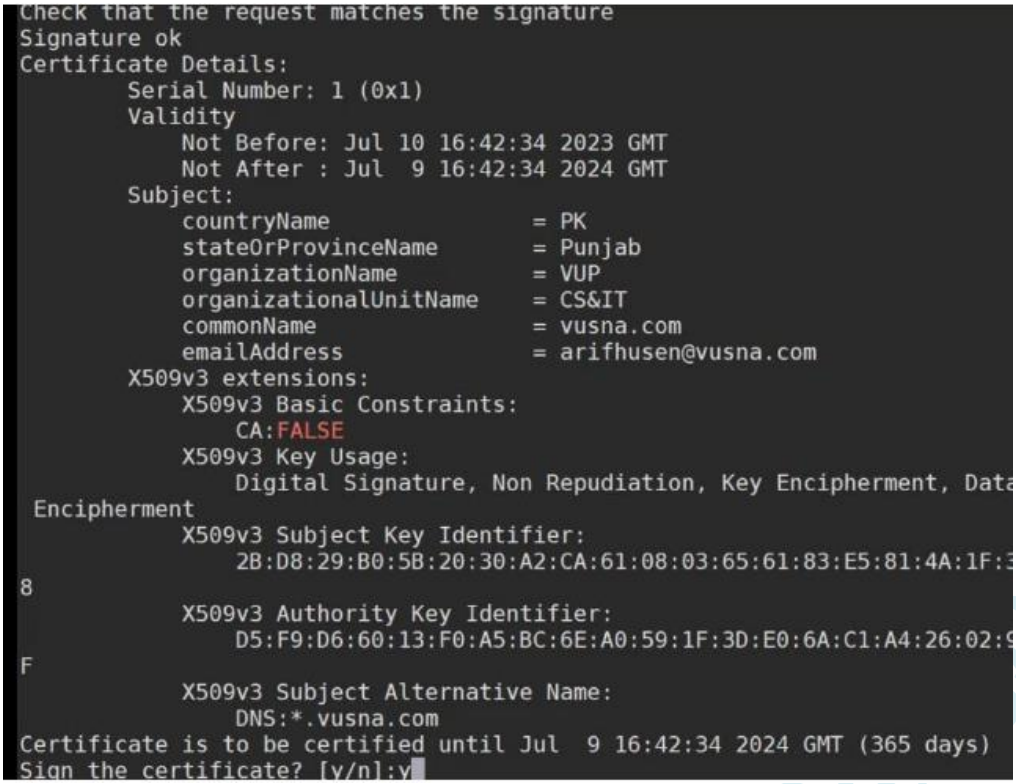
11) To generate it's csr request
   - root@ca:/etc/ssl/vusnaCA# openssl req -w -key /etc/ssl/vusnaCA/private/vusna.com.key -out /etc/ssl/vusnaCA/csr/vusna.com.csr -config /etc/ssl/vusnaCA/openssl.conf

our csr request is generated

12) To generate request to sign certificate and generate tls certificate
   - root@ca:/etc/ssl/vusnaCA# openssl ca-config /etc/ssl/vusnaCA/openssl.cnf -in /etc/ssl/vusnaCA/csr/vusna.com.csr -out /etc/ssl/vusnaCA/certs/vusna. com.crt

```
root@ca:/etc/ssl/vusnaCA# openssl genrsa -out /etc/ssl/vusnaCA/private/vus
na.com.key 4096
root@ca:/etc/ssl/vusnaCA# openssl req -new -key /etc/ssl/vusnaCA/private/v
usna.com.key -out /etc/ssl/vusnaCA/csr/vusna.com.csr -config /etc/ssl/vusn
aCA/openssl.cnf
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN
.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
PK [PK]:
Punjab [Lahore]:
Lahore []:
Virtual University of Pakistan [vusna]:
Computer Science and Information Technology []:
vusna.com []:
arifhusen@vusna.com []:

Please enter the following 'extra' attributes
to be sent with your certificate request
A challenge password []:abcd1234
An optional company name []:VUP
root@ca:/etc/ssl/vusnaCA# clear
```



LAB 13

In today's labs ham nfs kernel server ko configure Kary gy and Kuch directories and exports ko set Kary gy

1) To install nfs kernel
   - root@nfs:/# apt install nfs-kernel-server

2) To set export
   - root@nfs:/# vi /etc/exports

3) NFS Exports
   - /nfs shares/public(rw, sync,no_subtree_check. 0,no_root_squash)

mahiabeen9786@gmail.com

4) To check k ham ny Kya Kya share kiya Howa hy
   - root@nfs:/# df -h

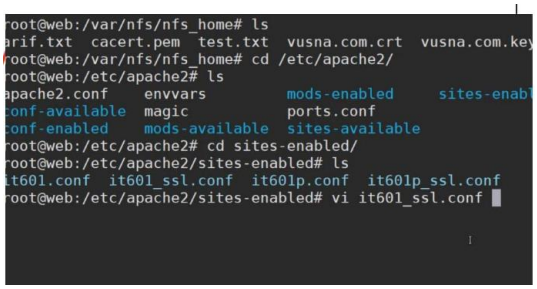
client side py ham nfs common ka option download karty hy

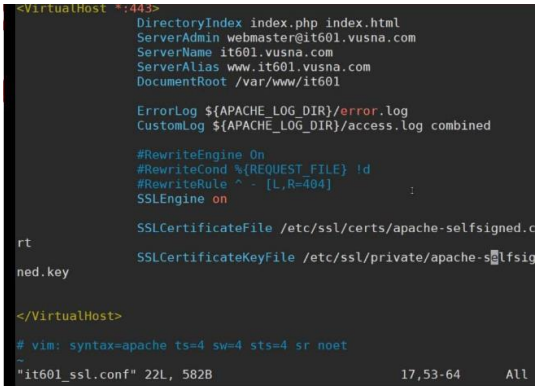
nfs client ko b ham apny web server py ja k configure karty hy

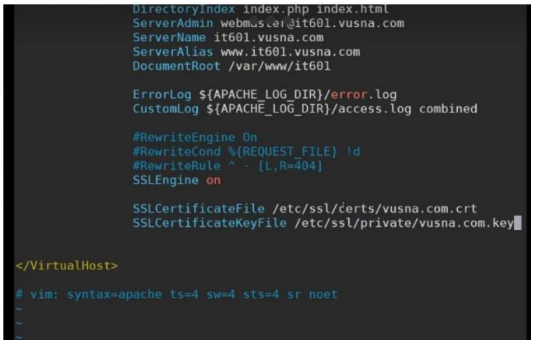
5) for nfs folder we have to create a directory.
   - root@web:/# cd/var/nfs
   - root@nfs:/var/nfs# cd nfs_home/

6) To share nfs share public directory to nfs home
   - root@web:/# mount -t nfs 192.168.19.6:/nfs_shares/public/var/nfs/nfs_home

Ab CA k certificate py jyy gy jinko ham ny pichali lab mai create kiya tha lakin ham nfs k sath connectd ni hy to pahly ham nfs ka client install kry gy







7) To install nfs client firstly create directory in var
   - root@ca:/# mkdir /var/nfs
   - root@ca:/# mkdir /var/nfs/nfs_home

8) To install nfs common for client
   - root@nfs:/# apt install nfs-common

9) To create mount point
   - root@ca:/# mount -t nfs 192.168.19.6:/nfs_shares/public/var/nfs/nfs_home

Now copy the crt file , private key and root certificate

mahjabeen19786@gmail.com

root@ca:/var/nfs/nfs_home# cp /etc/ssl/vusnaCA/certs/vusna.com.crt.

root@ca:/var/nfs/nfs_home# cp /etc/ssl/vusnaCA/private/vusna.com.key

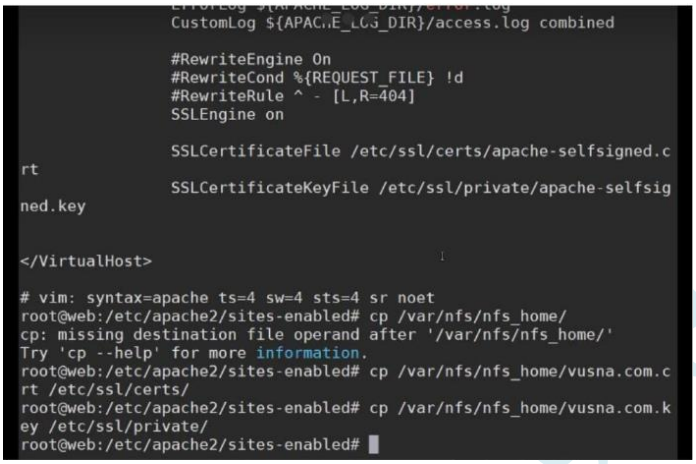
root@ca:/var/nfs/nfs_home# cp /etc/ssl/vusnaCA/cacert.pem

Now going to apache2 and it's enable sites

root@web:/var/nts/nfs home# cd /etc/apache2/

root@web:/etc/apache2# cd sites-enabled/

open i601_ssl.conf with vi command



Now copy all configuration here

now change some configuration

10) To restart the apache2 service
    - root@web:/etc/apache2/sites-enabled# systemctl restart apache2

11) To move the . htaccess file ta k jab ham access Kary http k thorough to hab sy password na pocha jyy
    - root@web:/var/www/it601# mv.htaccess old_htaccess.txt

mahjabeen369@gmail.com

lab 13 9 (PART 2)

Now in this lab we configure LDAP client ki authentication ko daikhy gy k ham osko kasy configure kar sakty user manager k Liya lam ka portal use Kary gy Wohi user authentic hongy

1) pahly ham server py login ho gy and then home ki directory Mai jayy gy
    - root@web_server:/# home/

wha pahly koi b directory ni honi chaiya

packages ko install Kary gy ab package ko install karny k Liya apt update kry gy repositories ko update k karny k Liya

Now we have to install three software packages

libnss-ldap

libpam-ldap

ldap-utils

2) Check the connectivity between client and server
    - root@web_server:/# ping ldap.vusna.com

3) To add parameter to make installation more interactive
    - root@web_server:/# echo 'debconf debconf/frontend select dialog' debconf-set-selections

4) To install packages
    - root@web_server:/# sudo apt -y install libnss-ldap libpam-ldap ldap-utils

Enter LDAP url

LDAP://ldap.vusna.com

DC=vusna,dc=com

can=admin,DC=vusna,DC=com

mahjabeen97869@gmail.com

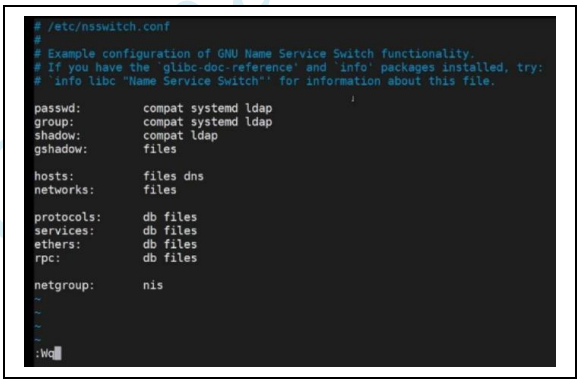
now add password and press enter

now we have to change some configuration in nsswitch

5) To open the nsswitch configuration
   - root@web_server:/# vi/etc/nsswitch_conf

6) After that Hamy common password ki configuration ko b edit Karna pry ga
   - root@web_server:/# vi /etc/pam.d/common-password

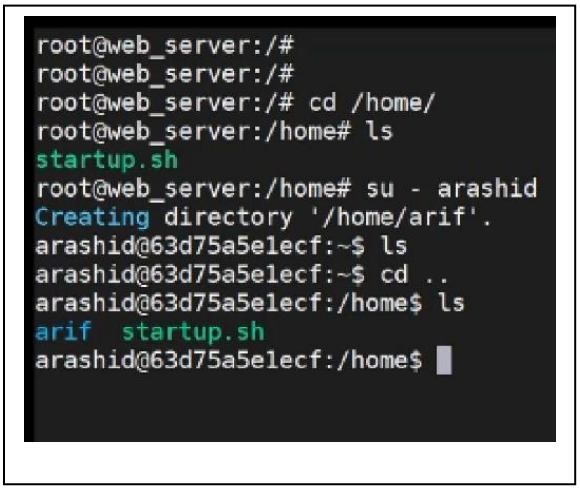
here on line 26 remove authtoken {authentication token}



7) To add information on common session
   - root@web_server:/# vi /etc/pam.d/common-session

session optional pam_mkhomedir.so skel=/ etc/skel umask=077

add this line ad the end and save the configuration



now install nscd service

root@web_server:/# apt install nscd

LAB 14

In today's lab ham email service ko configure Kary gy jo basically Kisi organization k internal email ki requirements ko fullfil Kary gi

mahjabeen978690@gmail.com

For email configuration we required some pre requisites that are

(dedicated server =root@email:/#)

Prerequisits:

1-DHCP Service

2- DNS Service

3-CA Service

4- NFS Service

Now we have to install these packages

1-SMTP Server 2 - IMAP/POP Server

Smtp is used to send mails and Imap is used to access external clients

1. To install postfix package for Ipam
    - root@email:/var# apt install postfix

wo ham sy pochy ga k apny is mail ko kis site k Tor py rakhna hy to ham 2 ka option select Kary gy (internet site) mean normal mail

now enter System mail name

System mail name: admingmail.vusna.com

installation is completed

2. To restart the postfix service
    - root@email:/var# systemctl restart postfix

3. To enable the postfix ta k jasy hi server start ho postfix ki service automatically enable ho jy
    - root@email:/var# systemctl enable postfix

mahjabeen97869@gmail.com

4. To check the version of postfix
    - root@email:/var# postconf mail_version

5. To check port 25 is enable or not or Kya ham is k connection enable kar sakty hy k ni kar sakty
    - root@email:/var# ss -lnpt | grep master

6. To check it is working on external host or not
    - we have to install telnet
    - root@email:/var# apt install telnet

7. To check it our port 25 is working fine and connected or not
    - root@email:/var# telnet email.vusna.com:25

8. To install mailutils jis sy ham email send and receive dono kar sakty
    - root@email:/var# apt install mailutils

9. To send email
    - root@email:/var# mail a FROM:admin@email.vusna.com admin@email.vusna.com

Cc: Subject: Test Email

i am testing the postfix smtp mail server

root@email:/var#

our emails received

10. To copy the main .cf file from etc directory of postfix
    - root@email:/etc/postfix #cp main.cf main.cf.bk

mahjabeen97869@gmail.com

11. To edit this main file
    - root@email:/etc/postfix #vi main.cf

```
host
relayhost =
mynetworks = 127.0.0.0/8 192.168.0.0/24 [::ffff:127.0.0.0]/104 [::1]/128
mailbox_size_limit = 0
recipient_delimiter = +
inet_interfaces = all
inet_protocols = all
                                                          43,39         Bot
```

To create aliases jinko ham user email k Liya use kar sakty mean ager os email py mail send ho gi to wo particular host user ko receive ho gi

- root@email:/etc/postfix # vi /etc/aliases

then change the postmaster to admin instead of root is k bad contact us ki ik email add Kary gy osko b admin kar dain gy fr isko save kar dain gy

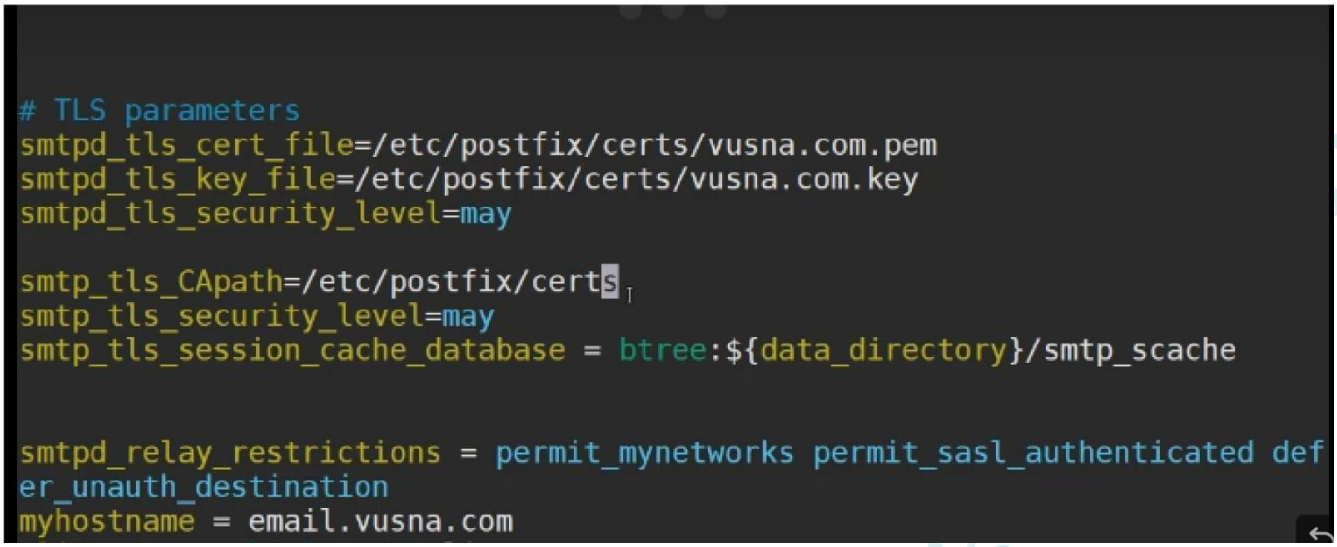
12. To create file in same directory
    - root@email:/etc/postfix # mkdir certs

13. Now ham is file mai apni shere directories ki files ko copy kar laity hy
    - root@email:/etc/postfix/certs # cp /var/nfs/nfs_home/* .

14. To update the postfix configurations
    - root@email:/etc/postfix # vi main.cf

mahjabeen97869@gmail.com

here configure tls configuration



15. To view the status all is okay or not
    - root@email:/etc/postfix # systemctl status postfix

LAB 14(part 2)

Ham ny basic postfix ki installation ko complete kar Liya hy ab ham ssl and tls ki different port ko open kar k on py postfix ko configure Kary gy

1. To check k abi configuration k doran Jo posrts ham ny enable ki hy wo open hoi Hain k nai
    - root@email:/etc/postfix # ss -lnpt grep master

Now we have to install dovecote IMAP and POP3 server

2. To install these packages
    - root@email:/etc/postfix # apt install dovecote-core dovecote-imapd dovecote-pop3

3. To check the version of dovecot
    - root@email:/etc/postfix # dovecot --version

mahjabeen97869@g...n

that is 2.3.16

4. To change the configuration of dovecot
   - root@email:/etc/postfix # cd/etc/dovecot/
   - root@email:/etc/devcot # vi dovecot.conf

protocols imapp

include try/usr/share/dovecot/protocols.d/*.protocol

5. To configure mail boxes locations
   - root@email:/etc/devcot#postconf mail_spool_directory

6. To specify the mail location in conf.d/
   - root@email:/etc/devcot/conf.d# vi 10-mail.conf
   - mail_location= maildir:~/Maildir

7. To add user in dovecot
   - root@email:/etc/devcot/conf.d# add user admin mail

8. To install packagedovecote.lmtpd
   - root@email:/etc/devcot/conf.d# apt install dovecote.lmtpd

vi k through is ki configuration Mai ja k protocol Mai lmptd b add kar dain gy

9. To open the master file
   - root@email:/etc/devcot/conf.d# vi 10-master.conf

mahjabeen97869@gmail.com

```
service pop3-login {
  inet_listener pop3 {
    #port = 110
  }
  inet_listener pop3s {
    #port = 995
    #ssl = yes
  }
}

service submission-login {
  inet_listener submission {
    #port = 587
  }
}

service lmtp {
  unix_listener /var/spool/postfix/private/dovecot-lmtp {
    mode = 0666
    user = postfix
    user = postfix
  }

  # Create inet listener only if you can't use the above UNIX socket
  #inet_listener lmtp {
    # Avoid making LMTP visible for the entire internet
```

Then main.cf Mai b ja k Kuch configuration ko change kar dy gy

10. To change the authentication in dovecot
    - root@email:/etc/devcot/conf.d# vi 10-auth.conf

Yaha plain text authentication mechanism ko disable kar dy gy

```
##
disable_plaintext_auth = yes

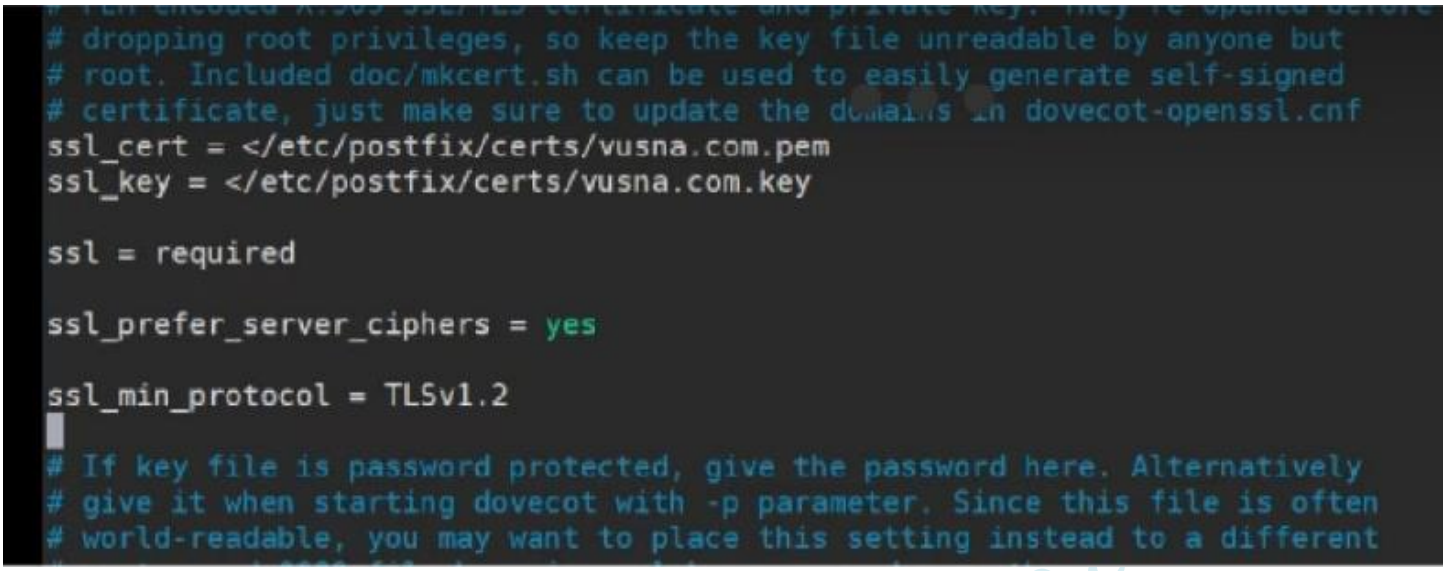
auth_username_format = %n

auth_mechanisms = plain login

# Disable LOGIN command and all other plaintext authentications unless
# SSL/TLS is used (LOGINDISABLED capability). Note that if the remote IP
```

11. To enable ssl configuration
    - root@email:/etc/devcot/conf.d# vi 10-ssl.conf

mahjabeen9@gmail.com



12. To edit the mail box configuration file
    - root@email:/etc/devcot/conf.d# vi 15-mailboxes.conf

And yha autocreate ko enable kar dain gy

13. To check the status of dovecot
    - root@email:/etc/devcot/conf.d# systemctl status dovecot

Then Microsoft outlook py email ko check kar lain

LAB 15

In today's lab ham log collection analysis and management k bary discuss kry gy

1. To add user elk
    - root@elk:/# adduser elk

set password

set name

2. To assign sudo permission to this user by adding them into sudo group
    - root@elk:/# adduser elk sudo

mahiabeen97869@gm

3. To switch from root to elk
    - root@elk:/# su elk
    - elk@elk:/$

Now we have to install java8

4. To install java8 firstly Install open jdk package
    - elk@elk:/$ sudo apt install openjdk-8-jdk

Now java is installed

5. To check the java version
    - elk@elk:~$ java - version

6. To install the elastic search tag firstly Install curl
    - elk@elk:/$ apt install curl

7. Ab ham Kuch switches use kar k elastic gpg ko download Kary gy
    - elk@elk:/$ curl -fsSL https://artifacts.elastic.co/GPG-KEY-elasticsea rch | sudo gpg-dearmor -o /usr/share/keyrings/elastic.gpg
8. To update repositories
    - elk@elk:/$ apt update repositories

Now we are able to install elastic search

9. To install elasticsearch
    - elk@elk:/$ sudo apt install elasticsearch

10. To reload daemon of elastic search
    - elk@elk:/$ sudo systemctl daemon-reload

11. To enable the elastic search service

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- elk@elk:/$ sudo systemctl enable elasticsearch.service

12. To change the configuration from etc directory
    - elk@elk:/etc$ sudo chmod 777 -R elasticsearch/

13. To open elastic search yml file
    - elk@elk:/etc/elasticsearch$ vi elasticsearch.yml

14. To check it on local host
    - elk@elk:/etc/elasticsearch$ curl -X GET 'http://localhost:9200'

15. To install kibana Jo Hamy ik dashboard service provide karta hy
    - elk@elk:/etc/elasticsearch$ sudo apt install kibana

16. To enable the kibana
    - elk@elk:/etc/elasticsearch$ sudo systemctl enable kibana

Ab isko start kar lain gy

configuration Mai server post 5601 hi rakhy gy

17. To configure name . configure file
    - root@dns:/etc/bind# vi named.conf

18. To set the permission
    - root@dns:/etc/bind# chmod 777 /var/log/bind/default.log

19. To make directory of bind
    - root@dns:/etc/bind# mkdir /var/log/bind
    - root@dns:/etc/bind# touch /var/log/bind/default.log
    - root@dns:/etc/bind# ls /var/log/bind/default.log

mahjabeen97869@gmail.com

LAB 15(part 2)

1. Now we have to install file beat deb package
   - root@dns:~# curl -L-0 https://artifacts.elastic.co/downloads/beats/filebeat/filebeat-8.8.2-amd64.deb

2. To install it with dpkg
   - root@dns:~# sudo dpkg -i filebeat-8.8.2-amd64.deb

3. To install log stash package
   - elk@elk:/etc/elasticsearch$ sudo apt install logstash

4. To go to logstash directory and open pipeline.yml
   - elk@elk:/etc/logstash$ vi pipelines.yml

pipeline.id: main

path.config: "/etc/logstash/conf.d/*.conf"

5. Go to conf.d file and open DNS configuration
   - elk@elk:/etc/logstash/conf.d$ vi 01-vudns.conf

Now we have to specify some information here

```
INPUT {
BEATS {
PORT=>5044
}
}
OUTPUT {
STDOUT{ }
}
```

mahjabeen97869@gmail.com

To run the logstash information kaya message Jo given configuration Mai thy wo run ho rahy k nai

- elk@elk:/usr/share/logstash/bin$ ./logstash -f /etc/logstash/conf.d/01-vudns.conf

6. To restart filebeat service
   - root@dns:/etc/filebeat# systemctl restart filebeat

7. To receive the data wee use dig command
   - root@dns:/etc/filebeat# dig web.vusna.com

Now ab hmary messages receive ho rahy hy

8. To adjust the data format
   - elk@elk:/usr/share/logstash/bin$ vi /etc/logstash/conf.d/01-vudns.conf

```
input {
beats {
port => 5044
type => "default
codec plain {
charset => "UTF-8"
}
}
}
filter {
grok {
patterns_dir => ["/etc/logstash/patterns/dns"]
match => { "message" => "%{BIND9_DATE: EventTime) queries: {LOGLEVEL: LogLevel):
client (DATA: ClientID} %{IP:ClientIP}[#]% (NUMBER: ClientPort} \({HOSTNAME:Hostname} \)[:] query:
{HOSTNAME: QueryName} %{WORD: QueryClass) %{WORD: QueryType) %(DATA:Query
```

mahjabeen97869@gmail.com

Flags) (%{IP: LocalAddress}\)"}

}

}

output {

stdout{

}

}

after saving the configuration run logstash

elk@elk:/usr/share/logstash/bin$ ./logstash -f /etc/logstash/conf.d/01-vudns.conf

9. To change the configuration of elastic search
   - elk@elk:/usr/share/logstash/bin$ vi /etc/logstash/conf.d/01-vudns.conf
   -

we have to specify elastic search host here

output {

stdout{

}

elasticsearch {

hosts => ["192.168.19.8:9200"]

index => "vusna-dns-%{+YYYY.MM.dd}"

}

10. To start the log stash and check k kaya Hamy data receive hota hy k nai hota
    - elk@elk:/usr/share/logstash/bin$ ./logstash -f /etc/logstash/conf.d/01-vudns.conf

now create the index pattern

Name= vusna-dns-*

Timestamp field= @timestamp

now our index is created

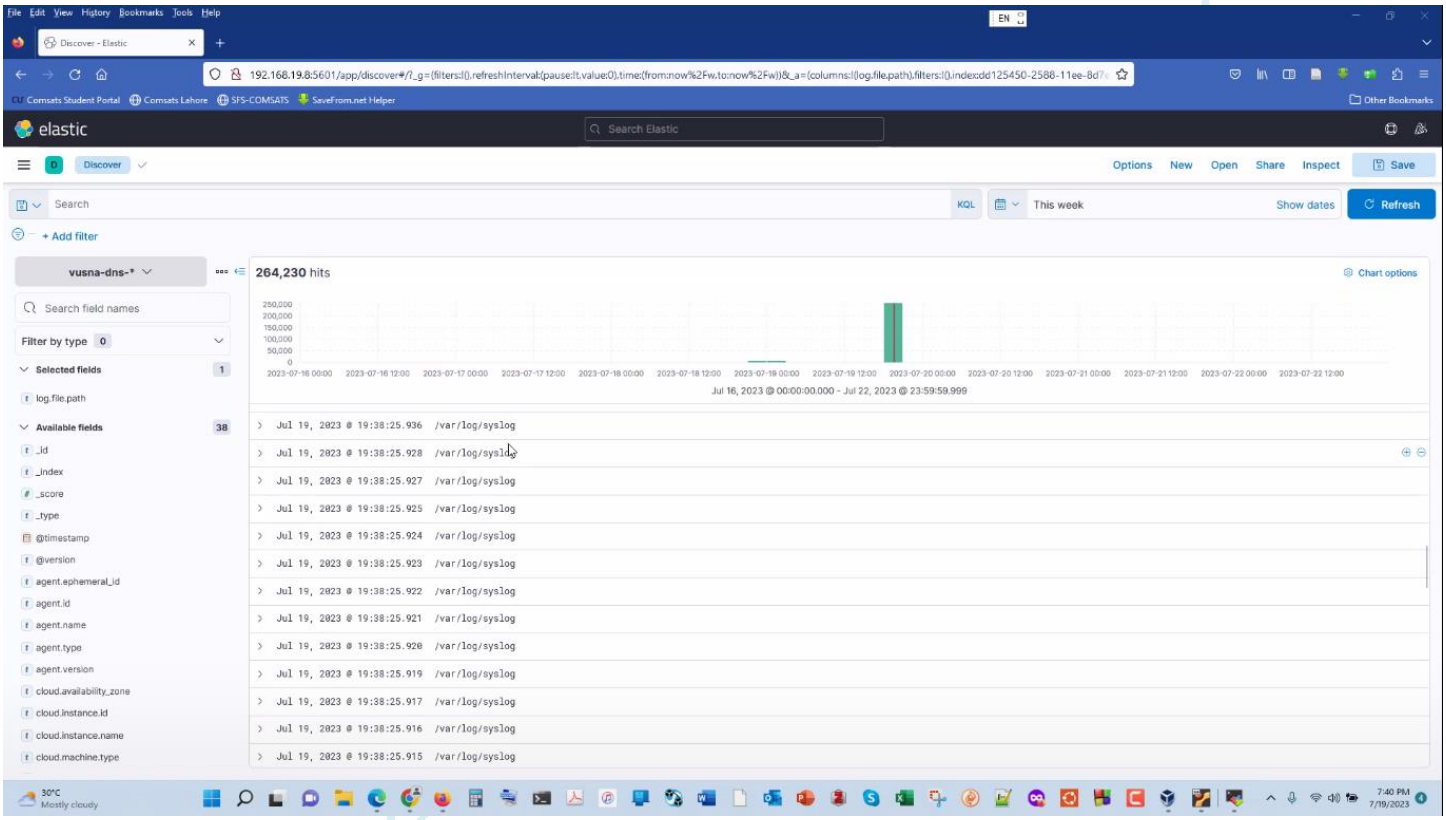
mahjabeen97869@gmail.com

LAB 16

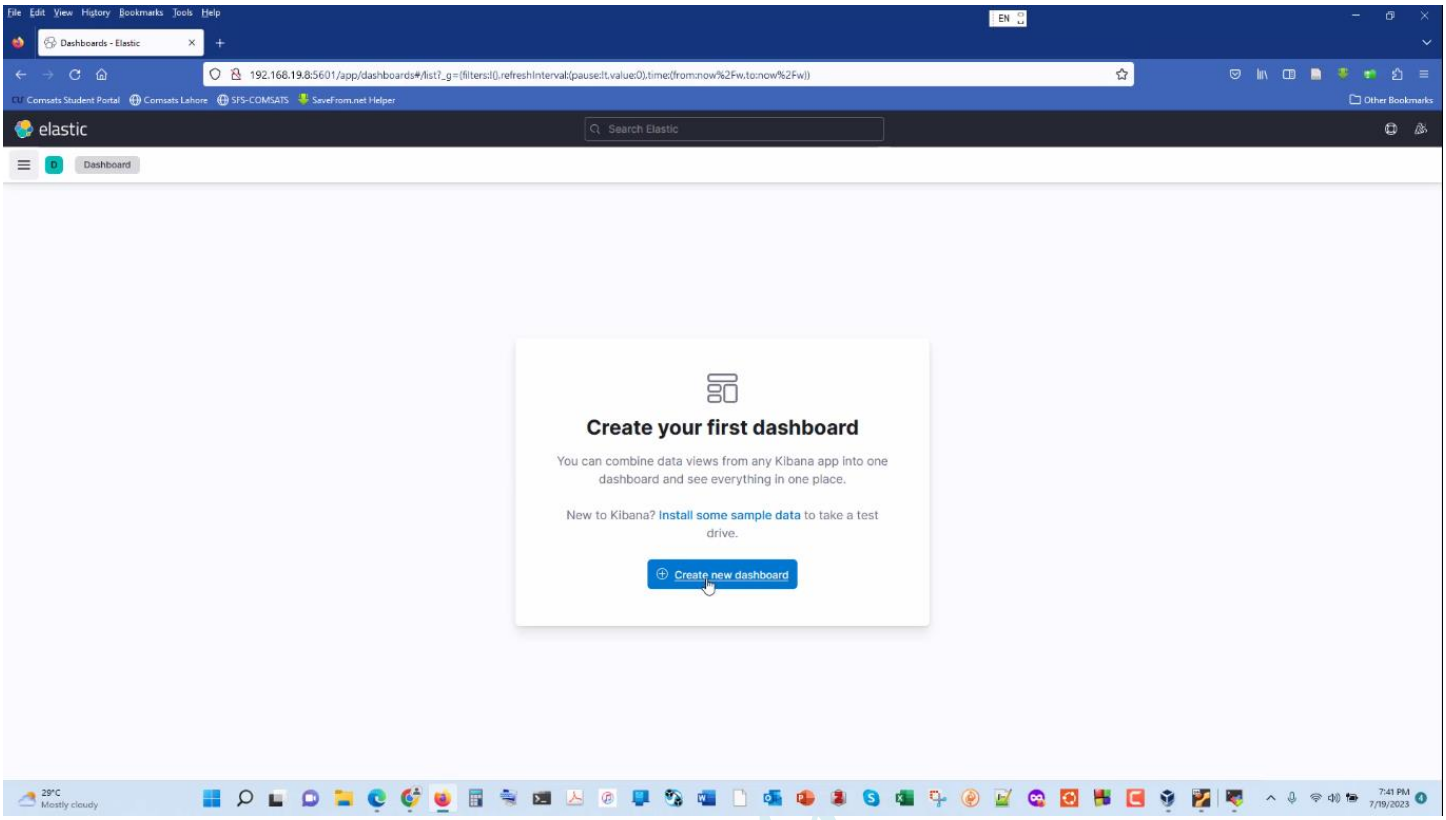
yha ham ny ik field Mai data show kiya tha ham multiple field ko show kar sakty ya onko khatam b kar sakty

log.file.path ko cross kar k onko multiple field Mai kar sakty

For create new dash board go to the option on left side three bars are given click the dash board and open it

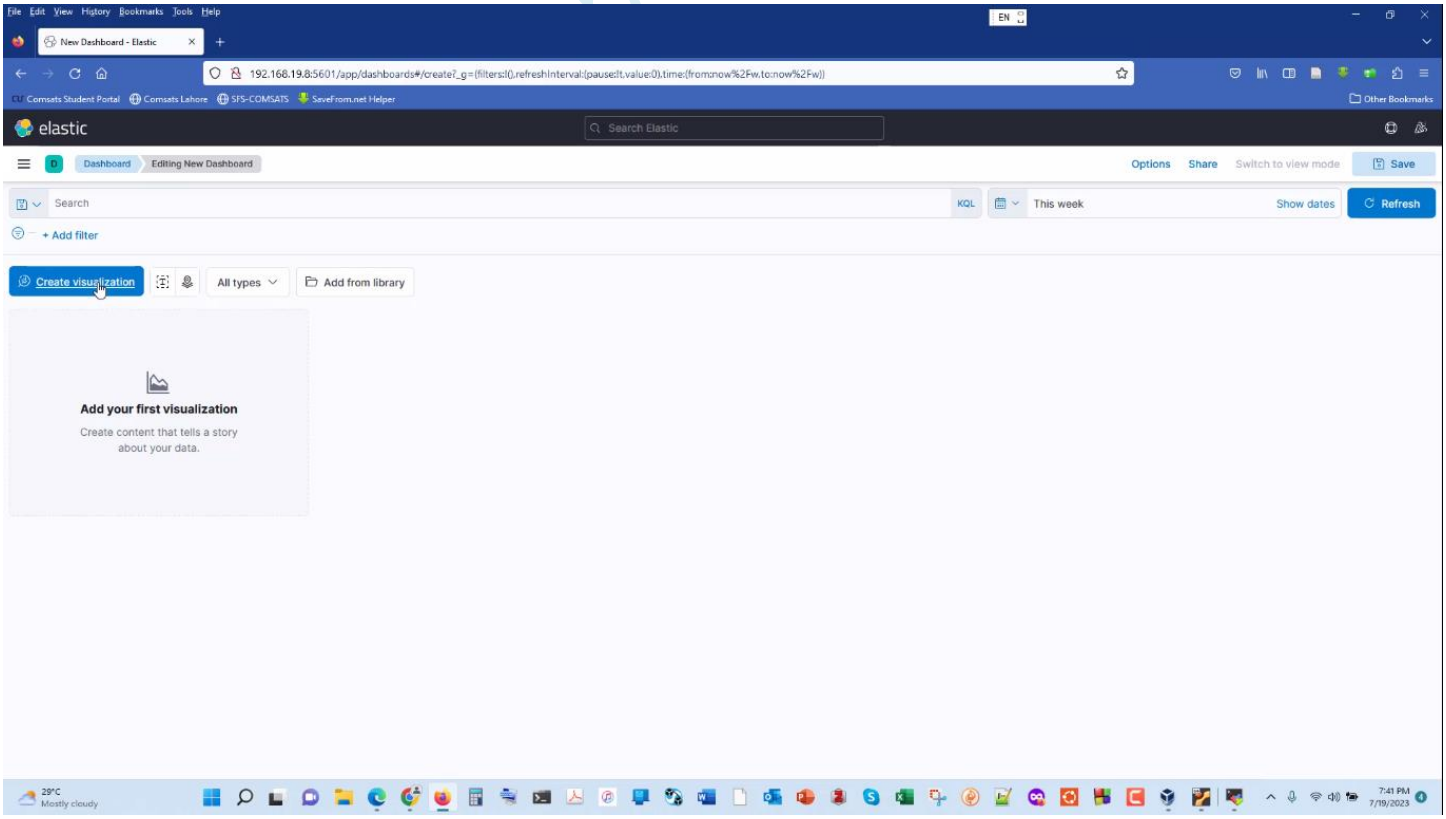


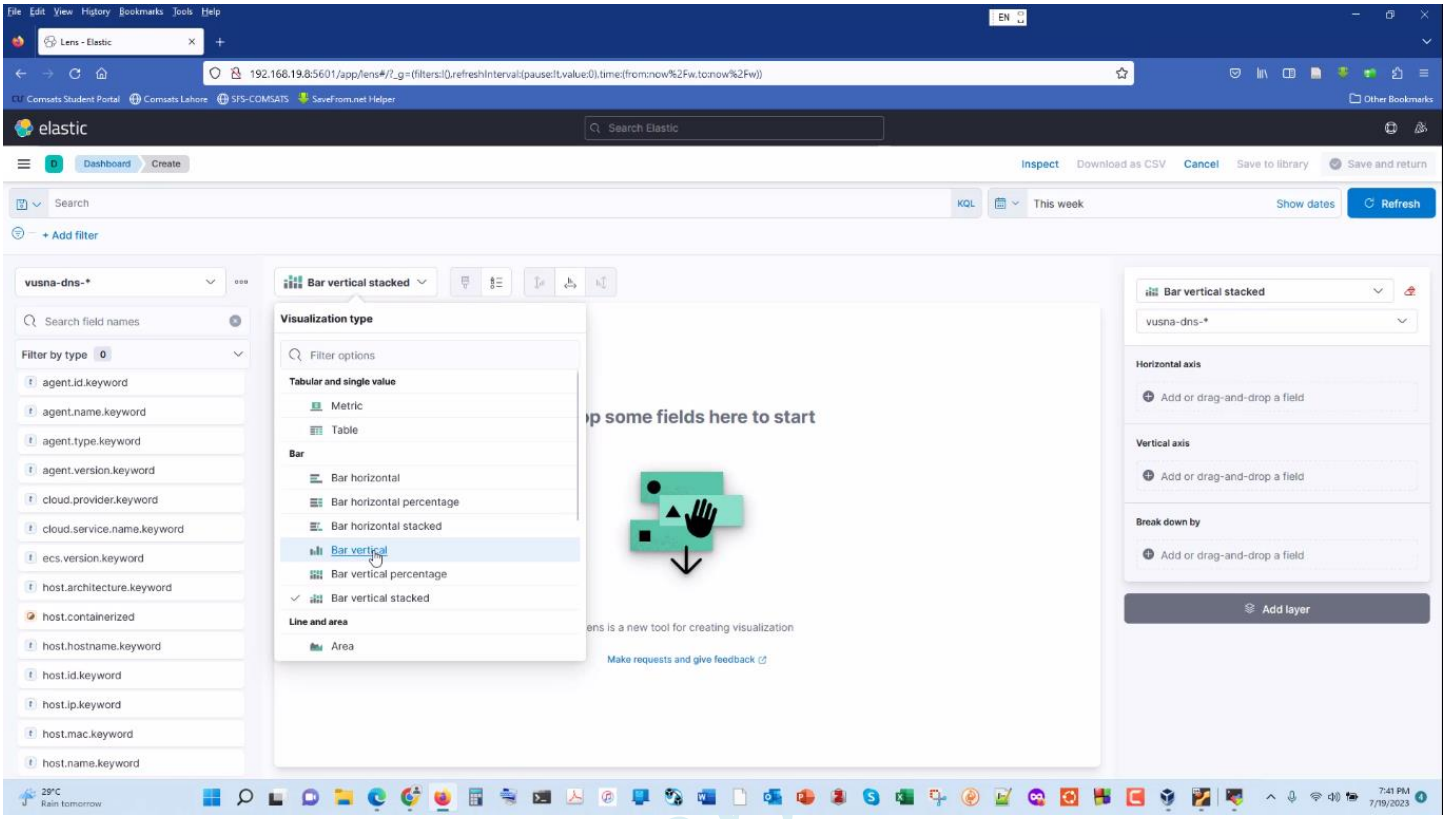
mahjabeen.com



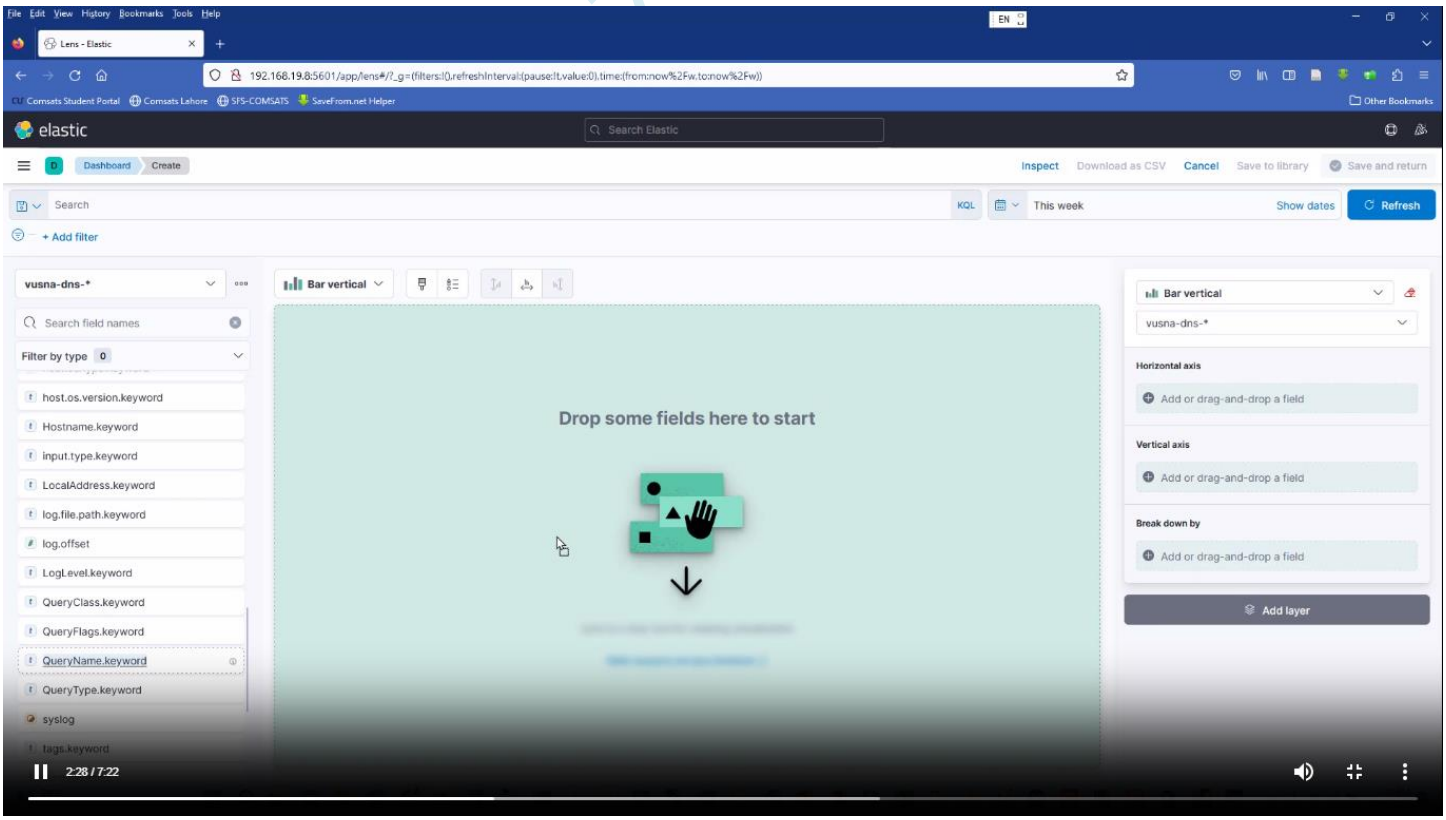
Go to create dashboard

field Mai Jo b field add karwana chaty wo add kar dy gy



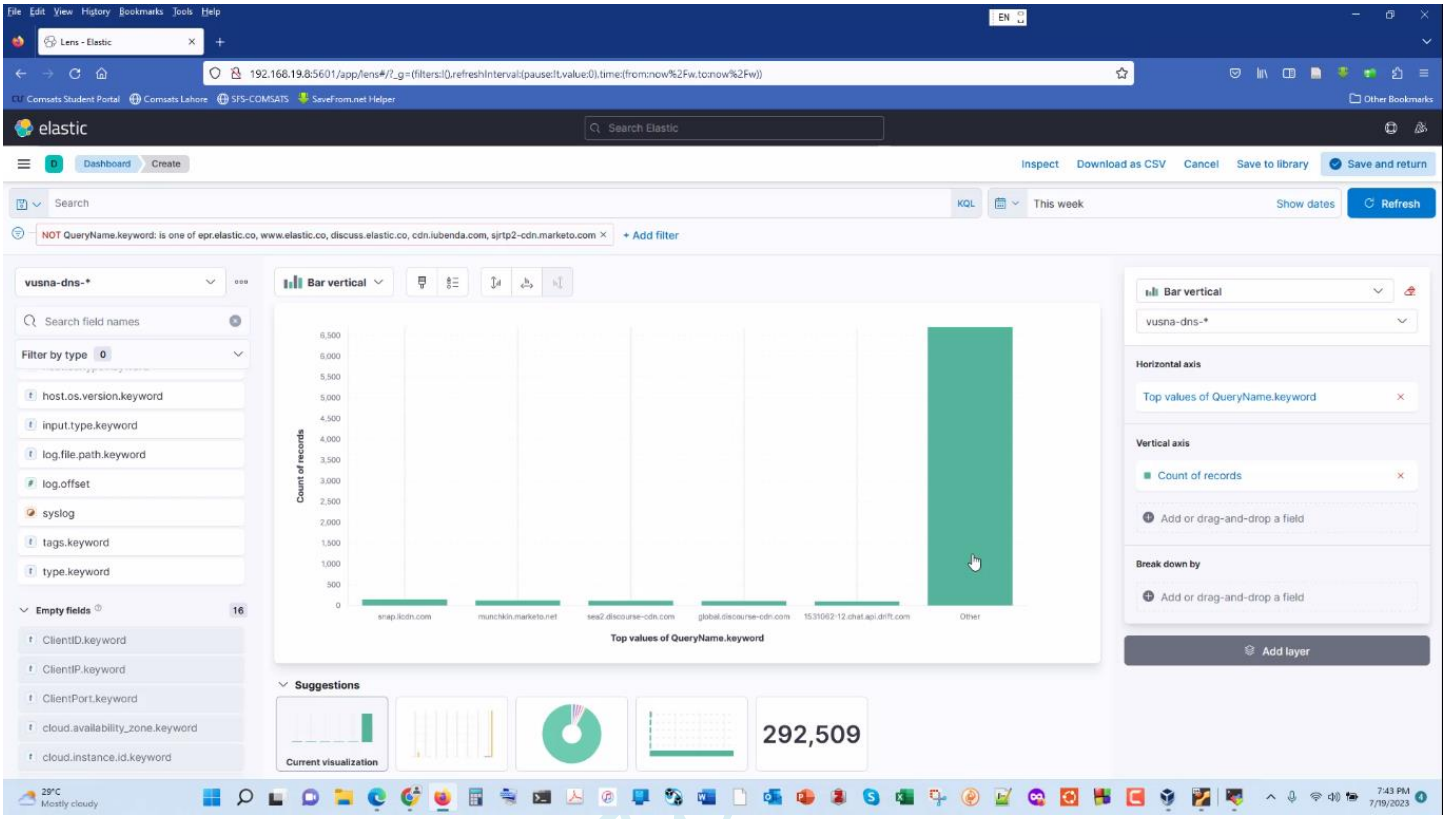


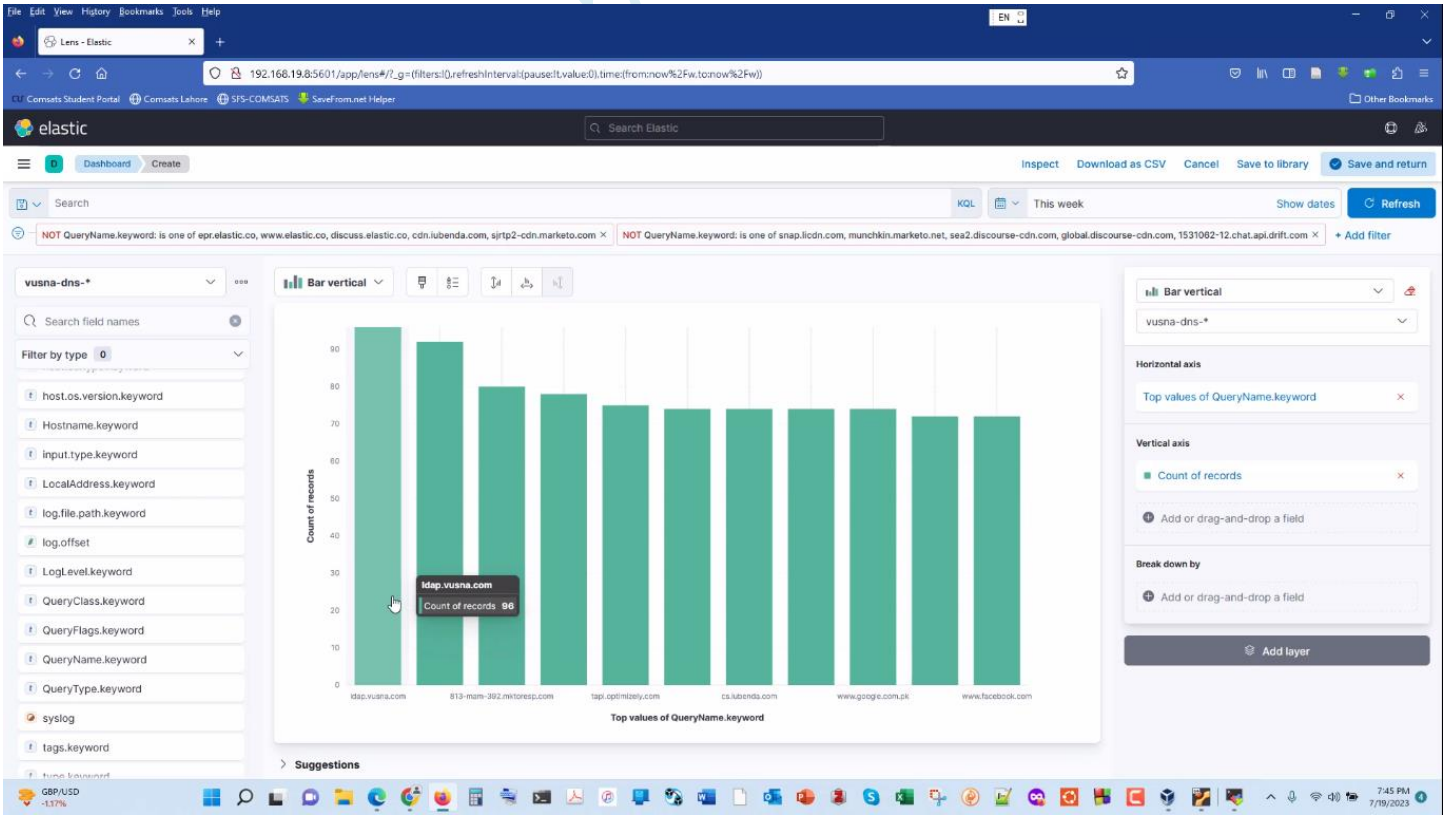
Now select the field (query name keyword) and drag it to the screen

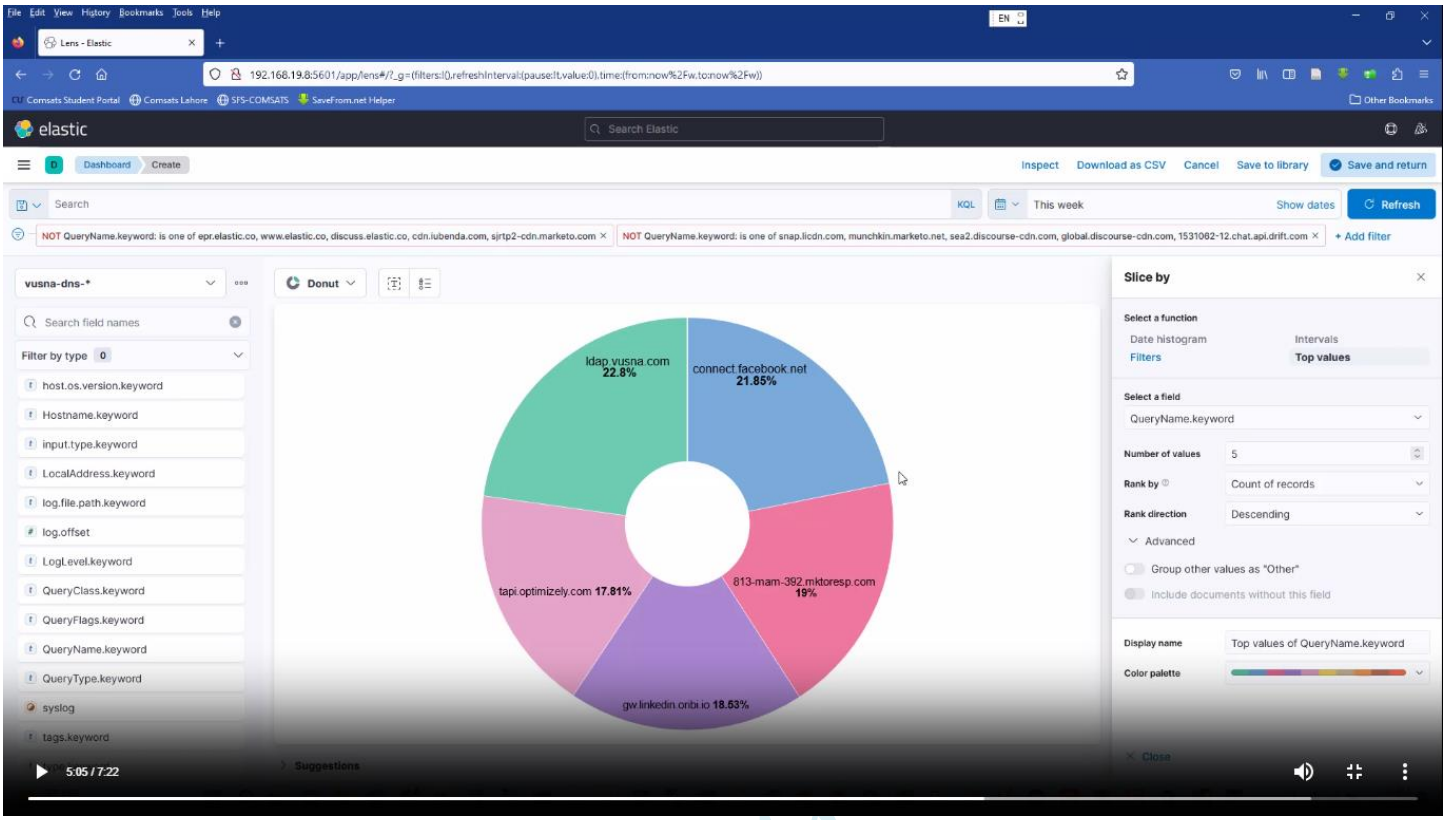


Change the shapes and form

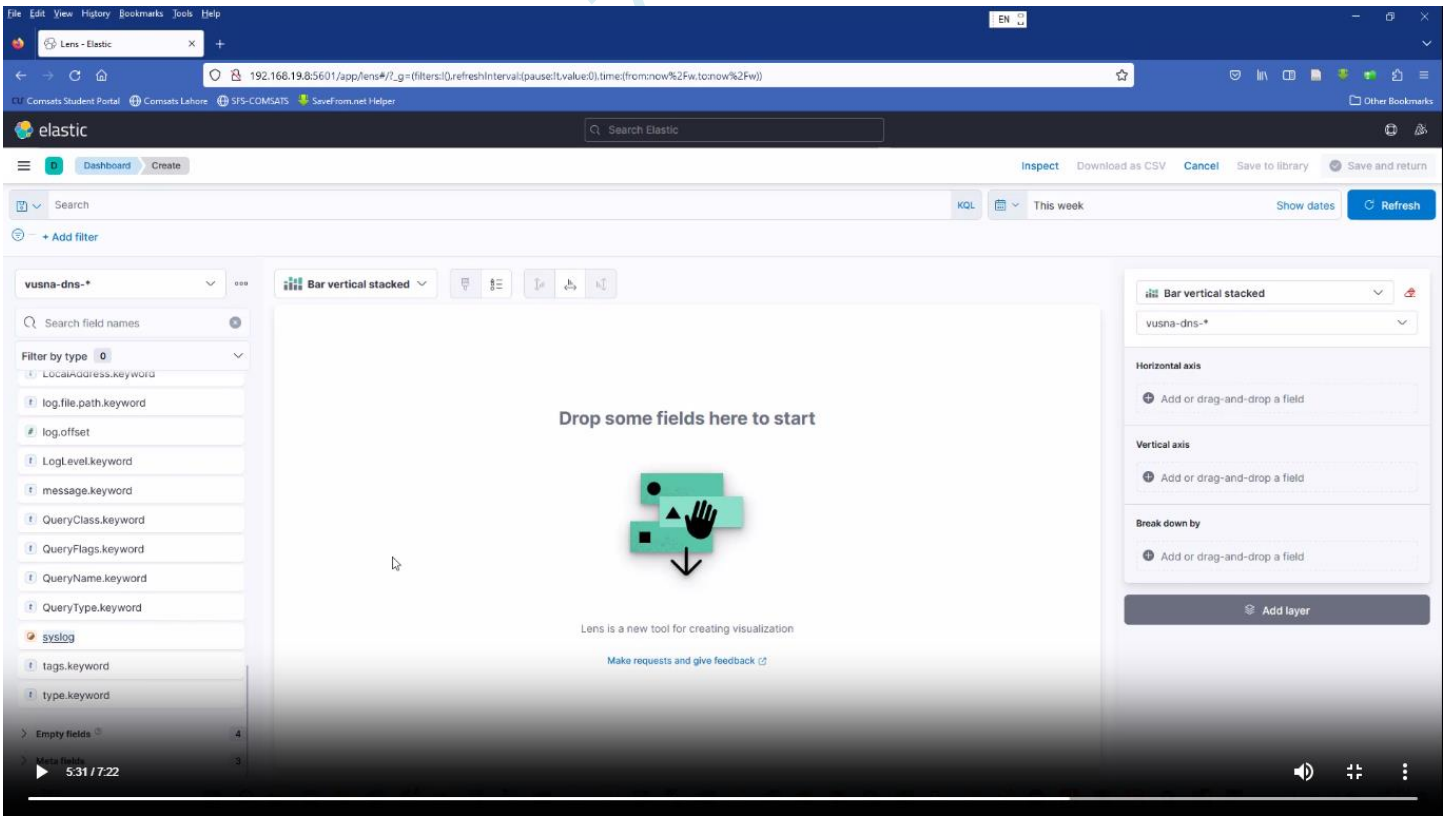
is bad sath hi graph banny lag jayy gy



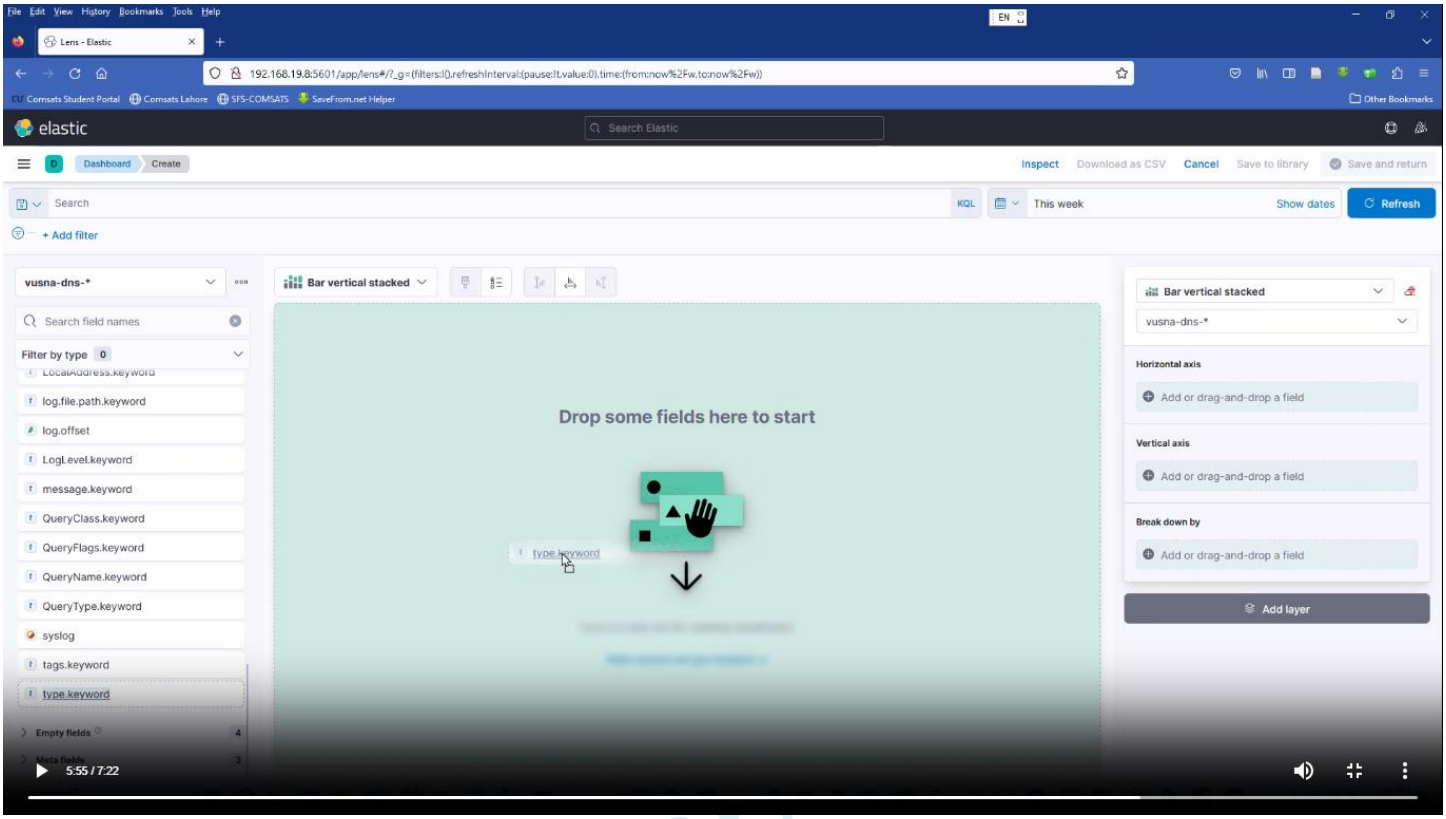




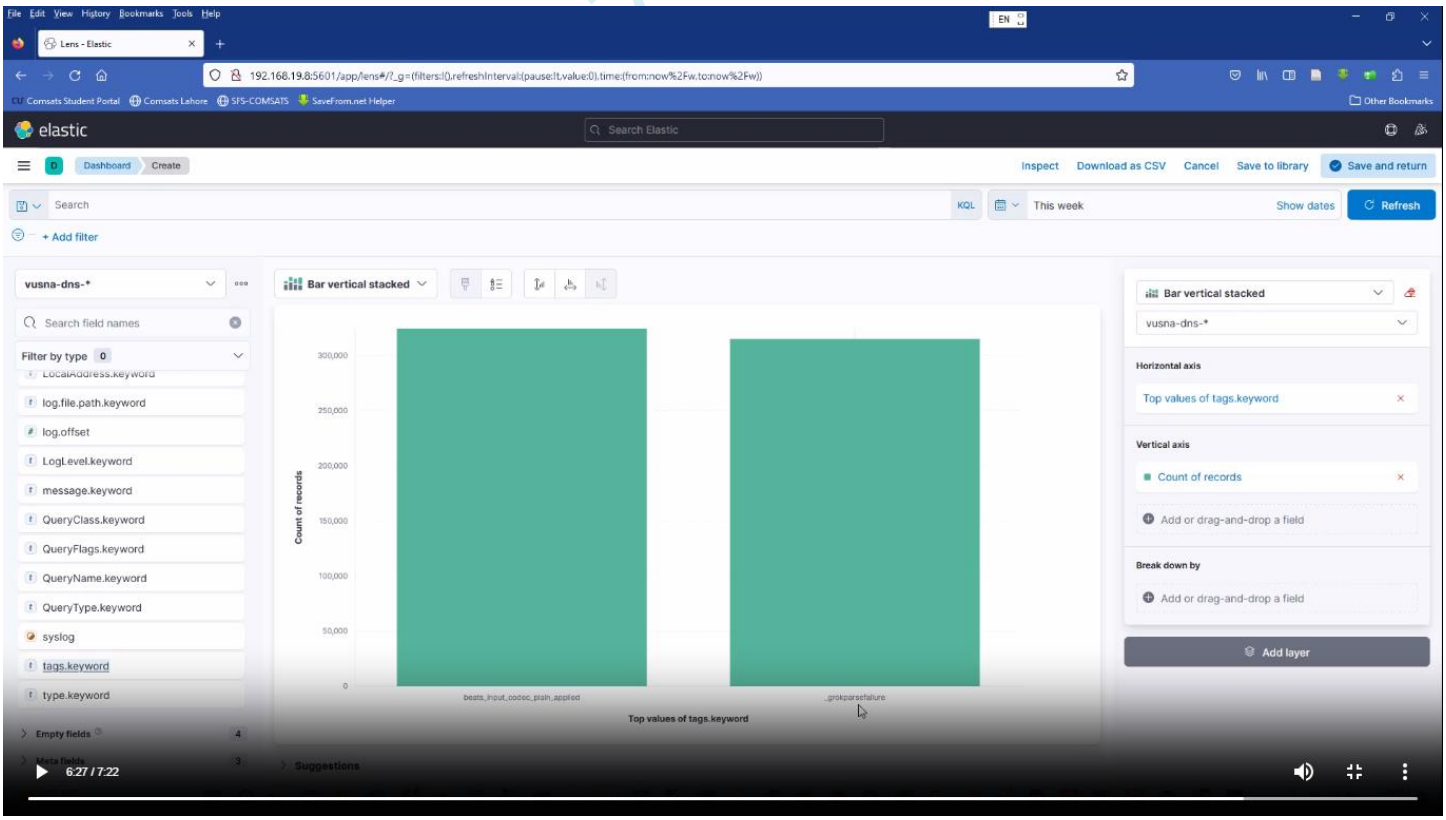
Save it and create an other syslog field

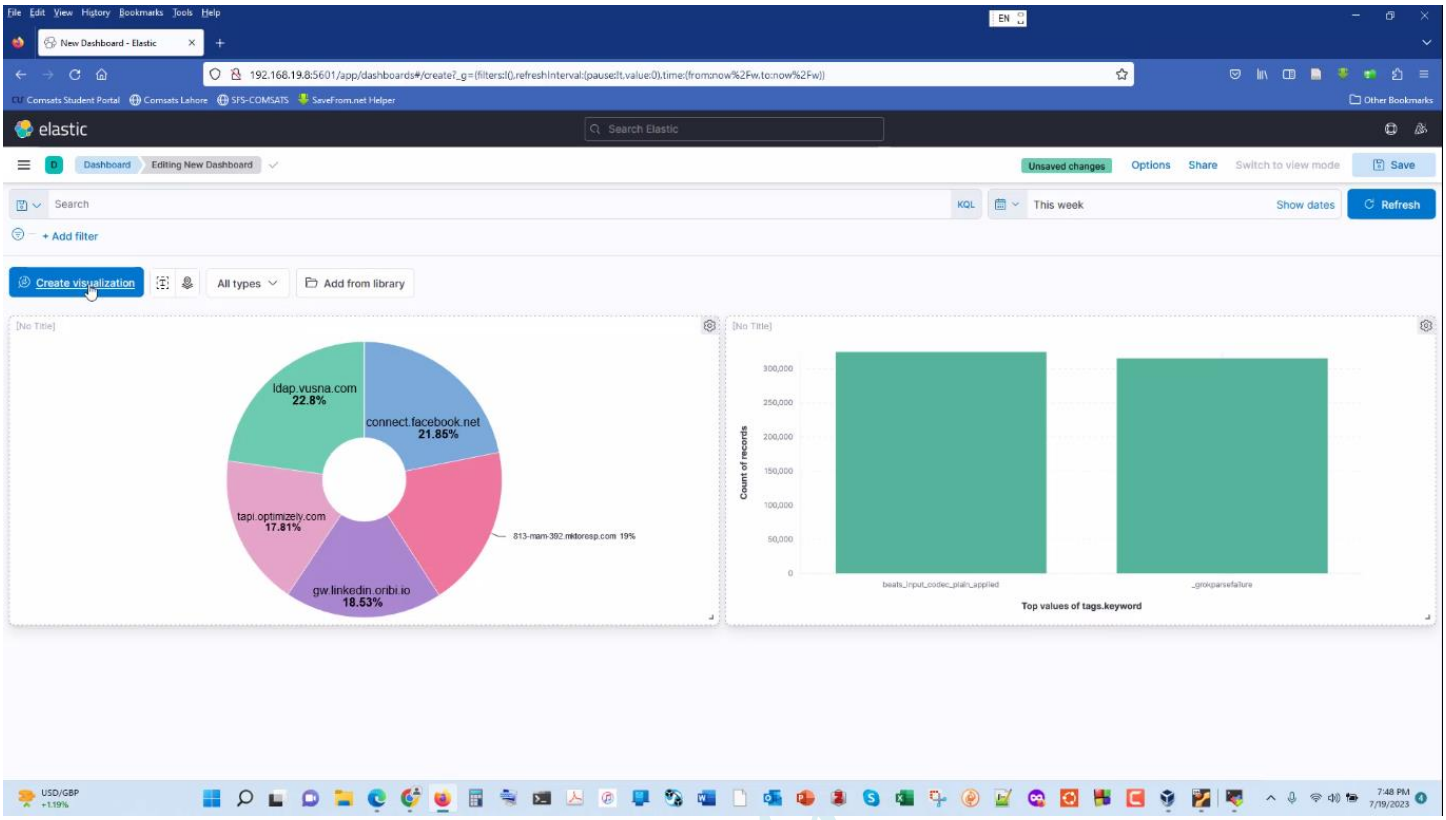


But q k ham ny syslog ko configure nai kiya tha is liya type ko select kary gy bas



Isi trah tag b add kar sakty





These are some of our dashboards

Machiene learning b ik important tool hy jis sy ham different tools ki analysis kar sakty hain

Best of luck for your studies

Remember me and my family in your prayers

JAZAK ALLAH

mahjabeen97869